Endogenous Education Choice in Segmented Search Markets

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Abstract

Structural models of education and the labour market are estimated in steady state or against business-cycle moments. Crises are neither: 2008 and 2020 for example were large, unanticipated shocks that plausibly shifted structural parameters—and the education margin registers the difference, with post-secondary training rising during the financial crisis but falling during COVID. This paper develops a general-equilibrium search model with two segmented labour markets in which the boundary between them moves in both directions: workers enter skill through an endogenous training cutoff, and trained workers may abandon the skilled market, rendering their training obsolete. I estimate the model on U.S. data as a pair of steady states around each crisis and compute the model-implied transition path between them. The financial crisis is measured as a demand shock that moves the economy along its Beveridge curve and pulls the training cutoff down; COVID as a matching-efficiency and reallocation shock that shifts the curve outward, compresses the skilled option value, and activates the cross-market exit—explaining the sign flip in training. Two applications use the estimated model directly. The first derives in closed form how outside options (b_U, b_S) move reservation match quality, unemployment duration, and match quality, and shows the two payoffs steer the education margins in opposite directions, so the right instrument depends on the crisis. The second compares education-subsidy designs: a training stipend and an upfront cost reduction are equivalent at the margin, but the stipend costs more per unit of enrolment response by a closed-form factor—a discounting wedge times a targeting wedge.

Important notice: the model is currently being re-estimated; existing tables and figures are placeholders from the previous estimation round and will be replaced in place (same file names). Interpretations of specific numbers are not to be taken seriously yet; [**TODO:**] marks in the text flag every claim to be confirmed against the new estimates.

1 Introduction

Structural models of education and the labour market are estimated in steady state or against business-cycle moments. Crises fit neither mould. The 2008 financial crisis and the COVID-19 pandemic were large, unanticipated shocks, and the labour markets they left behind differed from the ones they hit—not along a mean-reverting cycle, but in ways that look like shifts in structural parameters.

The education margin makes the point sharply. During the financial crisis the share of the U.S. working-age population enrolled in post-secondary training rose from 2.0 to 2.2 percent—the familiar countercyclical enrolment pattern documented across earlier downturns (Dellas and Sakellaris, 2003; Clark, 2011; Barr and Turner, 2015). During COVID it *fell*, from 3.0 to 2.9 percent, and the monthly hazard of entering training from unskilled unemployment fell by 17 percent. Two crises, the same direction of unemployment, opposite education responses. A single mechanism—training when the opportunity cost is low—cannot generate both.

This paper treats a crisis as what the data suggest it is: an unanticipated, permanent shift in a subset of structural parameters. I develop an equilibrium search model of education choice, estimate it as a *pair* of steady states around each crisis—deep parameters held fixed across the pair, regime-specific parameters re-estimated within each window—and then compute the model-implied transition path between the two equilibria. The estimation answers *which* parameters moved in each crisis and by how much. The transition answers *how* the economy carries its inherited composition of workers into the new regime, and at what speed each margin adjusts. Two policy applications then use the estimated model directly: the effect of non-employment payoffs on unemployment duration and match quality, and the design of education subsidies.

The model's core treats education as a market-access decision rather than a productivity upgrade. Untrained and trained workers search in separate frictional markets with qualitatively different internal dynamics. In the unskilled market, matches start at frontier quality and can only be damaged by idiosyncratic shocks; a worker's only route to a better match runs through unemployment. In the skilled market, match quality is drawn from a distribution, employed workers search on the job, and careers climb a ladder. A worker contemplating training therefore compares two entire continuation values—each shaped by its market's tightness, separation rates, and wage setting—and both continuation values are equilibrium objects. The boundary between the markets, the training cutoff $\bar{x}$, is determined jointly with pool compositions and tightness in both sectors through firms' free-entry conditions.

A second margin completes the model: trained workers can give up on the skilled market. A skilled unemployed worker may redirect her search to the unskilled market, and if she accepts a job there, her training becomes obsolete—she is bargained with, paid, and (after any future separation) re-enters unemployment as an untrained worker. This cross-market option binds for the lowest-ability trained workers, characterised by a second cutoff $\underline{x}$. In tranquil times the margin is dormant. It activates when the skilled market's matching capacity deteriorates, and it is what allows the model to generate *falling* training in a crisis without a counterfactual rise in the training cutoff: the marginal education investments of the past are undone first, and the skilled segment drains faster than demographic turnover alone would permit.

The estimation uses U.S. data from the CPS, ASEC, JOLTS, and Census Job-to-Job Flows over four windows: a pre-crisis baseline and a crisis window for each episode (2003–07 and 2008–10; 2015–19 and 2020–22). Within each window the model is estimated by simulated method of moments on eighteen moments—stocks, transition rates, wage moments, and market tightness by skill group. Deep parameters (the ability distribution and the non-employment flow payoffs) are estimated once per baseline and held fixed across the corresponding crisis window; everything else—productivities, matching efficiencies and elasticities, vacancy costs, shock arrival rates, match-quality distributions, bargaining weights, the training cost—is allowed to move. Comparing deep estimates across the two baselines, twelve years apart, provides a structural stability check on the classification. This design lets the data, rather than the model's a priori structure, decide what kind of shock each crisis was.

The two crises receive different answers. In the data, the financial crisis is a ride along the Beveridge curve: unemployment rises by three-fifths in both skill segments, tightness halves in both, job-finding rates fall by a quarter, and separation rates rise by a third—symmetric, demand-shock signatures. The estimated parameter shifts concentrate in productivities and vacancy costs, with the matching technology stable. The training cutoff falls and training rises: the model reproduces countercyclical enrolment through the compressed value of unskilled search. COVID looks different in every one of these dimensions. Separation rates jump far more for skilled workers; job-finding rates barely move; tightness *rises* in both segments; and the Beveridge curve shifts outward and stays there. The estimation maps this into a fall in matching efficiencies, a spike in the skilled shock arrival rate, and a deterioration of the skilled offer distribution. The skilled option value compresses for low-ability trained workers, $\underline{x}$ rises, the cross-market drain activates, and training falls.[1]

[1]Estimation of the current model version is in progress; the parameter tables and transition figures will be

The first policy application studies outside options. I derive in closed form how the reservation match quality responds to the non-employment payoffs b_U and b_S, and decompose the expected unemployment duration into a selectivity channel (workers reject more offers) and a market channel (tightness and contact rates respond through free entry), with expected match quality given by the truncated mean above the reservation cutoff. The decomposition gives structural content to the evidence in Nekoei and Weber (2017) that longer benefits improve match quality. The model adds a cross-market result with no single-market analogue: b_U and b_S pull both education margins in opposite directions. A more generous unskilled payoff raises $\bar{x}$ and $\underline{x}$—it discourages training and accelerates skill abandonment—while a more generous skilled payoff lowers both. The two instruments are not interchangeable, and which one mitigates a crisis depends on which margin the crisis attacks.

The second application compares two education subsidies that target the same margin: a flow stipend during training (b_T) and a reduction in the up-front training cost (c). At the margin the two are equivalent—a stipend increase moves the cutoff exactly as much as an up-front subsidy equal to its expected discounted value over the training spell. Away from the margin they differ, and the difference has a closed form: per unit of cutoff movement, the stipend's fiscal flow cost exceeds the cost subsidy's by the factor $\frac{r+\phi+\nu}{\phi+\nu} \cdot c(\bar{x})/\bar{c}_e$, where $\bar{c}_e$ is the entry-weighted average training cost. Both terms exceed one: the government pays stipends undiscounted while workers value them discounted, and stipends pay inframarginal trainees—who would train anyway—as much as marginal ones, while a proportional cost subsidy pays most where costs are highest, at the margin. General-equilibrium congestion in the skilled market, absent from competitive treatments of education subsidies (Heckman et al., 1998; Abbott et al., 2019), then determines how much of either subsidy survives.

The paper contributes, first, a search model in which the boundary between two frictional markets is an equilibrium object that can move in both directions—into skill through training, out of it through obsolescence; second, an estimation design that treats crises as regime pairs rather than business-cycle realisations, with an internal stability check and computed transitions between estimated steady states; and third, an analytical policy toolkit—reservation-response formulas, a duration–quality decomposition, and a subsidy-cost ranking—that follows from the estimated structure rather than sitting beside it. Section 2 positions the paper; Sections 3–4 present the model and the measurement; Section 5 reads the two crises through the estimates; Section 6 contains the applications.

updated in place, and the crisis-specific statements above should be read as the model-predicted configuration pending the final estimates.

2 Related Literature

This paper connects five bodies of work.

Education choice in search models

The most direct predecessors embed education decisions in search-and-matching frameworks. Charlot and Decreuse (2005) build a two-sector model in which workers self-select into educated and uneducated pools; both markets clear simultaneously, and a composition externality arises because the marginal entrant lowers the mean ability of the pool she joins, depressing vacancy creation for her co-searchers—so much so that too many workers educate in equilibrium. Charlot et al. (2005) study education as a source of both adaptability and productivity in a single frictional market. Charlot and Decreuse (2010) extend the two-sector framework to two-dimensional heterogeneity—ability and schooling cost—and show that the rich over-invest in education while the poor under-invest. Charlot and Malherbet (2013) add employment protection and find that, although firing costs can prop up education investments, the first-best instrument is a direct education subsidy. Wasmer (2006) studies the choice between general and specific skills in a single Mortensen–Pissarides market with firing costs.

These papers share two limitations that this paper removes. First, their skilled market has no internal career dynamics: once a worker enters the educated pool, wages follow from Nash bargaining and the match persists until destruction. Here the skilled market has on-the-job search and a job ladder à la Cahuc et al. (2006), so the value of training depends on the entire trajectory of climbing, not on an entry wage. Second, education in these models is an absorbing state. Here trained workers can abandon the skilled market, and their training becomes obsolete when they do—a margin that turns out to carry the COVID episode.

Search and matching foundations

The model builds on Diamond (1982) and Pissarides (2000), with endogenous job destruction from Mortensen and Pissarides (1994) governing the unskilled sector and the job-ladder structure of Postel-Vinay and Robin (2002) governing the skilled one; Cahuc et al. (2006) provide the wage-determination protocol. Robin (2011) adds aggregate shocks to the job-ladder model and shows that endogenous destruction coupled with worker heterogeneity amplifies them; Lise et al. (2016) and Lise and Robin (2017) estimate sorting models with two-sided heterogeneity; Moscarini and Postel-Vinay (2024) study how on-the-job search propagates business cycles through the ladder. This literature evaluates its models in steady state or against business-cycle moments—recurring, mean-reverting fluctuations, often a stochastic process for aggregate productivity. The premise of this paper is that 2008 and 2020 are not draws from that process: each is an unanticipated, persistent reconfiguration of several parameters at once, including, in 2020, the matching technology itself. Accordingly, where Lise and Robin (2017) take the type distributions in each market as given primitives, here the composition of both markets is endogenous, and the estimation is designed around regime pairs rather than cycles.

Acemoglu (2001) studies a mechanism related to my policy analysis in a one-market setting: when good and bad jobs coexist, higher unemployment benefits shift the composition of employment toward good jobs by raising reservation values. The cross-market analysis of Section 6 extends this logic to a setting where the composition response operates through education margins—which jobs workers train for, and which trained workers give up—rather than through firms' job-creation mix alone.

Human capital in search models

A separate literature studies skill accumulation within a given search market. Bagger et al. (2014) let human capital accumulate along the job ladder; Flinn et al. (2016) model firm-provided training decisions; Audoly et al. (2022) quantify how job-ladder losses, skill depreciation, and firm-specific capital combine in the earnings cost of displacement; Acabbi et al. (2024) connect human-capital ladders to cyclical sorting and hysteresis. In all of these, training changes a worker's productivity or position within one frictional environment. This paper takes the complementary view: training is a one-time, costly decision that changes *which market* the worker searches in, so the relevant return is a comparison of two continuation values governed by different frictions. The model deliberately holds ability fixed during non-employment; the panel evidence of Cohen et al. (2025), who find no decline in a broad battery of cognitive and non-cognitive skills over unemployment spells, supports this abstraction. Allowing experience-driven skill growth on the job remains a natural extension.

Education policy in general equilibrium

The competitive general-equilibrium literature on education policy—Heckman et al. (1998) and Abbott et al. (2019) most prominently—evaluates subsidies in OLG settings where the operative channel is the skill price: more graduates compress the premium. Bárány (2016) adds endogenous technology choice to this architecture. In my model the operative channel is market tightness: more trainees enter skilled unemployment, vacancy creation responds through free entry, and the contact rate facing every skilled searcher moves. The policy section also distinguishes a flow stipend from an up-front cost reduction—a design dimension on which Lindner and Reizer (2020) provide quasi-experimental evidence: frontloading Hungarian UI benefits shortened non-employment by a week and a half while leaving reemployment wages and job durations unchanged, holding total transfers roughly constant. The timing of a transfer, not only its size, moves behaviour. Card et al. (2018), synthesising over 200 active-labour-market-programme evaluations, find that training programmes show small short-run effects but larger gains two to three years out—the time profile a job ladder produces when newly trained workers enter at the bottom and climb.

Nekoei and Weber (2017) show, with an age-discontinuity in Austrian UI rules, that a nine-week benefit extension raises reemployment wages by about half a percent, and interpret the finding through two offsetting margins: selectivity (workers hold out for better matches) and duration (longer spells erode prospects). Section 6 derives both margins in closed form within the model and adds the cross-market margin that single-market frameworks cannot express: non-employment payoffs steer not only how long workers search but which market they search in.

Cyclical enrolment

An empirical literature documents countercyclical schooling. Dellas and Sakellaris (2003) establish the pattern in U.S. CPS data covering 1968–88: a one-percentage-point rise in unemployment raises college enrolment by about two percent. Clark (2011) finds large effects of youth labour-market conditions on post-compulsory enrolment in English regional panel data. Barr and Turner (2015) show that Great Recession enrolment responses were concentrated in two-year institutions and—uniquely in this literature—that expected UI benefit durations themselves raised unemployed workers' propensity to enrol. Blom et al. (2021) find that cohorts exposed to recessions while in college reallocate toward higher-paying majors. Ulvestad and Skjelbred (2023) extend the evidence to Norwegian master's programmes, where a one-point rise in field-specific unemployment raises applications by 6.5 percentage points. Bičáková et al. (2021, 2023) show that graduates who enrol in bad times earn more later, through greater effort in college rather than selection—a channel outside this model, which abstracts from the training spell's content.

This evidence base is built on demand-driven downturns, and the model agrees with it there: a demand shock compresses the value of unskilled search and pulls the training cutoff down. The new fact this paper documents and explains is the COVID exception—training *fell* while unemployment rose—which the model attributes to a different kind of shock operating on a different margin: not the value of entering skill, but the value of staying in it.

3 Model

I develop a general-equilibrium search model in which workers choose whether to invest in training and thereby gain access to a higher-skill labour market. Training decisions, wages, unemployment, and market tightness are determined jointly in both markets, so the education margin responds not only to the skilled wage premium but also to the availability of skilled jobs after training. The education-choice structure draws on Charlot and Decreuse (2005, 2010); the skilled-market dynamics build on the job-ladder framework of Postel-Vinay and Robin (2002) with the Nash-bargaining wage protocol of Cahuc et al. (2006); the unskilled market follows the endogenous-destruction tradition of Mortensen and Pissarides (1994). Two margins are new relative to this literature. First, the boundary between the two markets—the training cutoff—is an equilibrium object. Second, trained workers may abandon the skilled market: a skilled unemployed worker can redirect her search to the unskilled market, shedding her training on acceptance. The first margin governs how crises reshape education incentives; the second is what allows training to fall in a reallocation crisis.

All parameters below are constants. The multi-regime structure used in estimation—re-estimating a subset of parameters window by window—is a measurement device and is described in Section 4; it does not appear in the model.

3.1 Environment

Time is continuous and agents discount the future at rate $r > 0$. A unit mass of workers populates the economy. Each worker has an innate type $x \in [0, 1]$, fixed over time, drawn from a Beta(a_ℓ, b_ℓ) distribution with CDF L and density ℓ. Workers exit the economy at Poisson rate $\nu > 0$ and are replaced one-for-one by newborn untrained workers drawn from ℓ, which keeps the population composition stationary.[2]

Workers participate in one of two labour markets—unskilled (U) or skilled (S)—and move from the first to the second by completing training. Firms are homogeneous in both markets, and jobs carry an idiosyncratic match quality $p \in (0, 1]$. Production is multiplicative in type

[2]Treating x as fixed during non-employment is consistent with the panel evidence of Cohen et al. (2025), who find no decline in a wide range of cognitive and non-cognitive skills over the course of unemployment spells.

and match quality:

$$\pi_U(x,p) = P_U\, x\, p, \qquad \pi_S(x,p) = x\, P_S(x)\, p = \gamma_{P_S}\, x^{\gamma_{P_S}}\, p. \tag{1}$$

The unskilled productivity shifter $P_U > 0$ is a constant. The skilled shifter is a function of type, $P_S(x) = \gamma_{P_S}\, x^{\gamma_{P_S}-1}$ with $\gamma_{P_S} \geq 1$—the density of a Beta$(\gamma_{P_S}, 1)$ distribution. Setting $\gamma_{P_S} = 1$ recovers a skill-neutral technology; $\gamma_{P_S} > 1$ makes the return to the skilled market increase in ability. This single parameter governs how strongly the skilled market favours high-x workers, and it plays a central role in the cross-market mechanism below.

Each market matches seekers and vacancies through its own Cobb–Douglas matching function

$$M_j = \mu_j\, \big(\widetilde{\mathcal{U}}_j\big)^{\eta_j} \mathcal{V}_j^{1-\eta_j}, \qquad \eta_j \in (0,1),\ j \in \{U, S\}, \tag{2}$$

where $\widetilde{\mathcal{U}}_j$ is the mass of effective seekers in market j (defined below, since both markets attract seekers beyond their own unemployed) and $\mathcal{V}_j$ is the mass of vacancies. Tightness is $\theta_j = \mathcal{V}_j / \widetilde{\mathcal{U}}_j$, vacancies fill at rate $q_j(\theta_j) = \mu_j \theta_j^{-\eta_j}$, and a seeker contacts a vacancy at rate $\theta_j q_j(\theta_j)$. I write $f_U \equiv \theta_U q_U(\theta_U)$ for the unskilled contact rate and $\kappa_S \equiv \theta_S q_S(\theta_S)$ for the skilled one. These two rates are the prices through which conditions in each market feed into education decisions.

3.2 Training

Only untrained workers in unskilled unemployment may start training. Doing so requires a one-time utility cost

$$c(x) = (1-x)\, e^{\,c-x}, \tag{3}$$

which declines in ability; the parameter c enters through exponentiation so that the cost coefficient e^c is positive for any $c \in \mathbb{R}$. Training completes at Poisson rate $\phi > 0$, at which point the worker enters skilled unemployment. Training changes market access only; it does not change x.

While out of work, workers receive flow payoffs by status: b_U in unskilled unemployment, b_T in training, and b_S in skilled unemployment. These are composites of home production, informal income, transfers, and the amenity value of non-employment, and they are deep parameters of the environment.[3]

[3]In the U.S., UI eligibility requires prior employment and expires after a fixed duration, so b_j should not be read as a benefit payment alone. The policy experiments of Section 6 vary the b_j as stand-ins for permanent changes in the generosity of the non-employment safety net.

The value of training satisfies

$$(r+\phi+\nu)\,T(x)=b_T+\phi\,U_S(x), \tag{4}$$

where $U_S(x)$ is the value of skilled unemployment. Equation (4) is the channel through which skilled-market conditions reach the education margin: anything that compresses U_S compresses T one-for-one up to the factor $\phi/(r+\phi+\nu)$.

An untrained unemployed worker chooses between searching and training:

$$U_U(x)=\max\big\{\,U_U^{\text{search}}(x),\ -c(x)+T(x)\,\big\}, \tag{5}$$

where the search value satisfies

$$(r+\nu)\,U_U^{\text{search}}(x)=b_U+f_U\left(E_U(x,1)-U_U^{\text{search}}(x)\right) \tag{6}$$

and $E_U(x,1)$ is the worker value of a fresh unskilled match. Because $c'(x)<0$ and $T(x)$ rises in x while the unskilled technology is only linear in x, the net gain from training, $\Delta_T(x)\equiv -c(x)+T(x)-U_U^{\text{search}}(x)$, is increasing in ability.

Proposition 1 (Training cutoff)**.** *Under the maintained functional forms, $\Delta_T(x)$ is increasing in x, so the optimal training policy is a cutoff rule: there exists $\bar{x}\in[0,1]$ with $\tau(x)=\mathbf{1}\{x>\bar{x}\}$ and $\Delta_T(\bar{x})=0$. Workers with $x>\bar{x}$ train upon entering unskilled unemployment; workers with $x\le\bar{x}$ search in the unskilled market.*

The cutoff $\bar{x}$ summarises the education margin. It moves with the skilled premium, with the arrival rate of skilled jobs after training, and with the value of the unskilled outside option—all equilibrium objects.

3.3 Unskilled Labour Market

Match quality and separation

New unskilled matches begin at the frontier, $p_0=1$: every hire starts at full efficiency and can only deteriorate. Damage shocks arrive at Poisson rate λ_U and redraw efficiency from $G(p)=p^{\alpha_U}$ on $[0,1]$ (equivalently, $p'=e^{-d}$ with $d\sim\text{Exp}(\alpha_U)$). The match dissolves whenever the redraw falls below a reservation quality $p_U^*(x)$ determined in equilibrium. A worker's only route to a better unskilled match is through unemployment.

Surplus, separation rule, and wages

Let $S_U(x,p) = [E_U(x,p) - U_U(x)] + J_U(x,p)$ denote total match surplus, where J_U is the firm value. Nash bargaining with worker share β_U splits the surplus, $E_U - U_U = \beta_U S_U$ and $J_U = (1-\beta_U) S_U$. Summing the worker and firm Bellman equations eliminates the wage:

$$(r+\nu+\lambda_U)\, S_U(x,p) = P_U\, x\, p - (r+\nu)\, U_U(x) + \lambda_U \int_{p_U^*(x)}^{1} S_U(x,p')\, dG(p'). \tag{7}$$

The reservation quality solves $S_U(x, p_U^*(x)) = 0$:

$$p_U^*(x) = \frac{(r+\nu)\, U_U(x) - \lambda_U \int_{p_U^*}^{1} S_U(x,p')\, dG(p')}{P_U\, x}. \tag{8}$$

Higher-ability workers tolerate lower match quality because flow output rises faster in p when x is large; a better outside option raises the threshold and makes fragile matches dissolve sooner. The Nash sharing rule delivers a wage linear in match quality,

$$w_U(x,p) = \beta_U\, P_U\, x\, p + (1-\beta_U)(r+\nu)\, U_U(x). \tag{9}$$

Vacancies and free entry

Posting an unskilled vacancy costs k_U per unit of time. The pool of unskilled seekers contains the untrained unemployed *and* the skilled unemployed who have chosen to search in this market (Section 3.5): the effective seeker density is $\widetilde{u}_U(x) = u_U(x) + d(x)\, u_S(x)$, with aggregate $\widetilde{\mathcal{U}}_U = \int_0^1 \widetilde{u}_U(x)\, dx$. Every contact produces a match at $p_0 = 1$ and—by the bargaining convention below—the same firm value $J_U(x,1)$ regardless of the seeker's origin. Free entry drives the vacancy value to zero:

$$\frac{k_U}{q_U(\theta_U)} = \frac{\int_0^1 J_U(x,1)\, \widetilde{u}_U(x)\, dx}{\widetilde{\mathcal{U}}_U}. \tag{10}$$

Because the training cutoff determines which types remain in the unskilled pool, and the cross-market choice determines which trained types re-enter it, both education margins feed directly into unskilled vacancy creation.

3.4 Skilled Labour Market

The skilled market features a job ladder: matches form at heterogeneous quality, employed workers may search on the job, and quits to better matches are an equilibrium outcome.

Wages follow Nash bargaining in the presence of on-the-job search, as in Cahuc et al. (2006).

Match quality and separation

A skilled match forms at a quality p drawn from $\Gamma = \mathrm{Beta}(a_\Gamma, b_\Gamma)$ on $[0,1]$. Quality shocks arrive at rate λ_S and redraw p from Γ; the match dissolves whenever the redraw falls below the reservation cutoff $p^*_S(x)$. There is no exogenous separation: all skilled separations are endogenous, the product of a quality shock and a redraw below $p^*_S(x)$, or demographic exit at rate ν. Unlike unskilled matches, skilled matches can improve after a shock—a feature that interacts with the job ladder to generate rich mobility.

On-the-job search

An employed skilled worker chooses whether to search, at flow cost $\sigma_S \geq 0$. A searcher receives outside offers at the same contact rate κ_S faced by the skilled unemployed, each an independent draw $\tilde{p} \sim \Gamma$, and quits if the offer beats the current match. Workers in weak matches search because the expected improvement is large; workers in strong matches do not. The search policy is a second cutoff $p^{oj}_S(x) \geq p^*_S(x)$: the worker searches if and only if $p < p^{oj}_S(x)$, with indifference, $E^1_S = E^0_S$, at the cutoff itself.

Surplus and wages

For each search regime $s \in \{0,1\}$, define $S^s_S(x,p) = [E^s_S(x,p) - U_S(x)] + J^s_S(x,p)$ and let $S_S = \max\{S^0_S, S^1_S\}$. Under Nash bargaining with worker share β_S, the wage-free surplus recursions are

$$(r + \nu + \lambda_S)\, S^0_S(x,p) = \gamma_{P_S} x^{\gamma_{P_S}} p - (r+\nu)\, U_S(x) + \lambda_S\, \mathcal{I}_S(x), \tag{11}$$

$$\begin{aligned}\big(r + \nu + \lambda_S + \kappa_S(1-\Gamma(p))\big)\, S^1_S(x,p) = {}& \gamma_{P_S} x^{\gamma_{P_S}} p - (r+\nu)\, U_S(x) - \sigma_S \\ &+ \lambda_S\, \mathcal{I}_S(x) + \kappa_S \beta_S \int_p^1 S_S(x,p')\, d\Gamma(p'),\end{aligned} \tag{12}$$

where

$$\mathcal{I}_S(x) \equiv \int_{p^*_S(x)}^1 S_S(x,p')\, d\Gamma(p') \tag{13}$$

is the skilled tail integral—the option value of a fresh draw from Γ. The poaching hazard $\kappa_S(1-\Gamma(p))$ appears on the left of (12): offers that beat p arrive at that rate and end the match from the firm's perspective. The separation cutoff solves $S_S(x, p^*_S(x)) = 0$. Derivations are in Appendix A.

The two wage schedules are

$$w_S^0(x,p) = \beta_S\,\gamma_{P_S} x^{\gamma_{P_S}} p + (1-\beta_S)(r+\nu)\,U_S(x), \tag{14}$$

$$w_S^1(x,p) = w_S^0(x,p) + (1-\beta_S)\,\sigma_S - \beta_S(1-\beta_S)\,\kappa_S \int_p^1 S_S(x,p')\,d\Gamma(p'). \tag{15}$$

On the search region the last term dominates, so searchers earn less than non-searchers at the same (x,p): bargaining shifts part of the search cost onto the worker, and the firm anticipates the shorter expected duration of a match whose worker is climbing the ladder. The discount is largest near the reservation threshold, where the ladder option is most valuable.

Vacancies and free entry

Posting a skilled vacancy costs k_S per unit of time. The seeker pool combines the skilled unemployed who search in this market and the employed searchers:

$$\widetilde{u}_S(x) = (1-d(x))\,u_S(x) + \int_0^1 s^*(x,p)\,e_S(x,p)\,dp, \qquad \widetilde{\mathcal{U}}_S = \int_0^1 \widetilde{u}_S(x)\,dx, \tag{16}$$

where $s^*(x,p) = \mathbf{1}\{p < p_S^{oj}(x)\}$ is the search policy and e_S the density of skilled employment. A vacancy's expected value at contact, $\mathcal{J}_S$, averages the firm surplus over this composition, counting only offers that are accepted (above $p_S^*(x)$ for the unemployed; above the current p for poached workers). Free entry requires

$$q_S(\theta_S)\,\mathcal{J}_S = k_S. \tag{17}$$

The training cutoff governs the inflow of trained workers, the OJS cutoff governs how many employed workers stay on the market, and the cross-market choice removes part of the unemployed pool—all three margins enter (17) through $\widetilde{u}_S$ and $\mathcal{J}_S$.

3.5 Cross-Market Search

A trained worker whose skilled prospects are poor need not wait for them to improve. Skilled unemployed workers choose, type by type, which market to search:

$$d(x) \in \{0,1\}, \qquad d(x) = 1 \iff \text{search the unskilled market.}$$

Search is exclusive: the worker is contacted at rate κ_S if $d(x) = 0$ and at rate f_U if $d(x) = 1$. A skilled worker who accepts an unskilled job sheds her training at that moment: the bargained wage uses $U_U(x)$ as her outside option, the firm receives the standard $J_U(x, 1)$, and upon any later separation she re-enters *untrained* unemployment—from which she may, if her type warrants it, pay $c(x)$ and train again. Training is obsolete once abandoned. This bookkeeping convention is what allows unskilled free entry (10) to use a single firm-value surface.

Conditional on the choice, the value of skilled unemployment takes one of two candidate forms:

$$U_S^{(0)}(x) = \frac{b_S + \kappa_S \beta_S \mathcal{I}_S(x)}{r + \nu}, \tag{18}$$

$$U_S^{(1)}(x) = \frac{b_S + f_U\, E_U(x, 1)}{r + \nu + f_U}, \tag{19}$$

and the worker picks the larger:

$$d(x) = \mathbf{1}\big\{U_S^{(1)}(x) > U_S^{(0)}(x)\big\}, \qquad U_S(x) = \max\big\{U_S^{(0)}(x),\, U_S^{(1)}(x)\big\}. \tag{20}$$

The flow payoff b_S is received in either case; only the search technology differs. The branch values make the trade-off transparent: staying in the skilled market buys the option value $\kappa_S \beta_S \mathcal{I}_S(x)$, which scales with $x^{\gamma_{P_S}}$; switching buys a faster, flatter payoff $f_U E_U(x, 1)$, which scales with x.

Proposition 2 (Search-market cutoff)**.** *Suppose $\mathcal{I}_S(x)$ is increasing in x (which holds for $\gamma_{P_S} > 1$ with an interior p_S^*). Then $U_S^{(1)}(x) - U_S^{(0)}(x)$ is decreasing in x, and there exists $\underline{x} \in [0, 1]$ such that $d(x) = 1$ for $x \le \underline{x}$ and $d(x) = 0$ for $x > \underline{x}$.*

Low-ability trained workers are the ones who abandon the skilled market: their option value there is convex in x and small, while the unskilled technology values them linearly. The cutoff $\underline{x}$ rises when the skilled market deteriorates—a fall in γ_{P_S}, a less favourable offer distribution Γ, or a fall in the matching efficiency μ_S all compress $\mathcal{I}_S(x)$—and falls back to zero when the unskilled market collapses too ($f_U \downarrow$), restoring the no-cross-market equilibrium. In tranquil times $\underline{x} = 0$ and the mechanism is dormant; it activates precisely in episodes that damage the skilled market's matching capacity. Section 5 shows that this is the COVID configuration.

3.6 Stationary Equilibrium

Let $u_U(x)$, $t(x)$, $u_S(x)$, and $e_S(x,p)$ denote the stationary densities of untrained unemployment, training, skilled unemployment, and skilled employment, and let $e_U(x)$ denote untrained employment, which pools original untrained workers with ex-trained workers absorbed through the cross-market channel. Flow balance pins these down; the full system is in Appendix A. Three relations carry the economics. The training stock balances entries against completions and exits,

$$t(x) = \frac{\tau(x)}{\phi + \nu}\, u_U(x). \tag{21}$$

The trained population balances completions against demographic exit and the cross-market drain,

$$m_S(x) = \frac{\phi\, t(x)}{\nu + d(x)\, f_U}, \tag{22}$$

so on the $d = 1$ region the skilled segment retains less of its inflow: trained workers leak back into unskilled employment at rate f_U. Finally, on the $d = 1$ region no skilled matches form, so $e_S(x,p) = 0$ and $u_S(x) = \phi t(x)/(\nu + f_U)$ there.

A *stationary equilibrium* is a collection of value functions $(U_U, U_U^{\text{search}}, T, E_U, U_S, E_S^0, E_S^1)$, firm values (J_U, J_S^0, J_S^1), policy rules $(\tau, d, p_U^*, p_S^*, p_S^{oj}, s^*)$, market tightnesses (θ_U, θ_S), and densities (u_U, t, u_S, e_S) such that:

(i) **Optimality.** Value functions satisfy their Bellman equations; the training rule, the search-market rule, the separation rules, and the OJS rule are optimal given those values.

(ii) **Wage setting.** Wages satisfy the Nash rules (9), (14), and (15).

(iii) **Free entry.** Tightnesses satisfy (10) and (17), each evaluated on its effective seeker pool.

(iv) **Stationarity.** Densities satisfy the flow-balance conditions, with (21)–(22) and the accounting identities $m_U(x) = \ell(x) - m_S(x)$, $e_U(x) = m_U(x) - u_U(x) - t(x)$.

The two markets are linked in both directions: the unskilled block supplies the skilled block with the trained mass m_S, the contact rate f_U, and the fresh-match value $E_U(\cdot, 1)$; the skilled block supplies the unskilled block with U_S and the cross-market seekers $d\, u_S$. I compute the equilibrium with a nested fixed-point algorithm, described in Appendix B.

3.7 Transition Dynamics

A crisis in this paper is an unexpected, permanent change in a subset of parameters at a date T. For $t > T$, value functions and policy rules are those implied by the new parameters—they jump at the instant of the switch—while the cross-sectional distributions are predetermined and evolve gradually under the new laws of motion:

$$\partial_t\, t(x,t) = \tau(x,t)\, u_U(x,t) - (\phi+\nu)\, t(x,t), \tag{23}$$

$$\partial_t\, u_U(x,t) = \nu\, \ell(x) + \delta_U(x,t)\, e_U(x,t) - \big(f_U(t) + \tau(x,t) + \nu\big)\, u_U(x,t), \tag{24}$$

$$\begin{aligned}\partial_t\, u_S(x,t) = {} & \phi\, t(x,t) + \delta_S(x,t)\, e_S^{\text{tot}}(x,t) \\ & - \Big(\nu + (1-d(x,t))\, \kappa_S(t)\big(1-\Gamma(p_S^*(x,t))\big) + d(x,t)\, f_U(t)\Big)\, u_S(x,t),\end{aligned} \tag{25}$$

with endogenous destruction hazards $\delta_U = \lambda_U G(p_U^*)$ and $\delta_S = \lambda_S \Gamma(p_S^*)$, and an analogous law for $e_S(x,p,t)$. Tightness satisfies free entry date by date, evaluated on the current pools. The trained mass obeys

$$\partial_t\, m_S(x,t) = \phi\, t(x,t) - \nu\, m_S(x,t) - d(x,t)\, f_U(t)\, u_S(x,t). \tag{26}$$

The transition path is the fixed point of a forward–backward iteration—values solved backward given tightness paths, distributions integrated forward given policies—detailed in Appendix B.

Three clocks govern adjustment speed. Values, policies, and tightness adjust within months: they are forward-looking objects disciplined by free entry. The training pipeline adjusts at rate $\phi + \nu$: a shift in $\bar{x}$ changes skilled-market inflows only as trainees complete. The skilled stock m_S adjusts at rate ν through demographic replacement—except where the cross-market drain is active, in which case the third term of (26) empties the skilled segment at rate $f_U \gg \nu$ on the affected types. The drain is therefore not a steady-state phenomenon but a transition accelerator: it lets the skilled share fall faster than demography alone would allow, exactly the asymmetry that distinguishes a reallocation crisis from a demand crisis in the data.

3.8 Model Properties

Four properties guide the empirical work. Proofs are in Appendix A.

Separation cutoffs. For each x there exist unique reservation qualities $p_U^*(x)$ and $p_S^*(x)$, and an OJS cutoff $p_S^{oj}(x) \geq p_S^*(x)$; all three are decreasing in x. Higher-ability workers hold on to worse matches because their flow output covers the outside option at lower p; they also stop paying the search cost at lower match qualities.

Two margins, one ranking. The training cutoff $\bar{x}$ and the search-market cutoff $\underline{x}$ order the trained population by the same index. Along a transition in which the drain activates, the workers who abandon the skilled market are those who trained under the old regime but fall below the new cross-market cutoff—the region $x \in (\bar{x}^{\,\text{pre}}, \underline{x}^{\,\text{post}}]$. These are the lowest types that ever entered: reallocation crises undo the *marginal* education investments first.

Composition channel in unskilled wages. When $\underline{x}$ rises, ex-trained workers—drawn from above the old training cutoff—enter unskilled employment. Average unskilled wages rise through composition alone, at fixed bargaining and productivity parameters. A one-market model reads this as wage growth; here it is reallocation.

Tightness responses. Free entry transmits parameter changes to tightness through four channels: directly through k_j and productivity; through the composition of the unemployed pools; through the matching technology (μ_j, η_j), which shifts the Beveridge curve itself; and mechanically through $d(\cdot)$, which inflates the unskilled seeker pool and deflates the skilled one. The third and fourth channels are the model's vocabulary for the COVID episode: an outward Beveridge shift with rising tightness and a draining skilled segment.

4 Data, Moments, and Estimation

4.1 Data and Windows

The model is estimated on U.S. data from five sources. Labour-market stocks, the training share, and worker transition rates come from the CPS Basic Monthly files. Wage moments come from the CPS Annual Social and Economic Supplement (ASEC). Vacancies come from JOLTS, allocated to skill segments through the CPS industry composition. The skilled employer-to-employer rate comes from the Census Job-to-Job Flows. Enrolment counts from the National Student Clearinghouse's IPEDS universe, combined with NCES time-to-degree figures, anchor the training-share level and the pre-estimated completion rate. Workers without a bachelor's degree are unskilled; workers with one are skilled. Construction details—sample restrictions, panel matching, the COVID misclassification correction, the JOLTS allocation—are in Appendix C.

Moments are computed over four windows, each treated as a stationary regime:

Table 1: Estimation windows

Label	Period	Length	Role
Pre-FC baseline	Jan 2003 – Dec 2007	60 months	Stage 1: all parameters
Financial crisis	Jan 2008 – Dec 2010	36 months	Stage 2: regime-specific only
Pre-COVID baseline	Jan 2015 – Dec 2019	60 months	Stage 1 + stability check
COVID crisis	Jan 2020 – Dec 2022	36 months	Stage 2: regime-specific only

The two baselines cover mature expansions; the two crisis windows cover each contraction and enough of its aftermath to capture where the labour market settled—through 2010, when tightness had not yet recovered its pre-2008 locus, and through 2022, when the Beveridge curve had not returned to its pre-2020 position. The symmetric design (60 months per baseline, 36 per crisis) puts the two crisis pairs on equal footing. Within each window I average monthly observations into a single moment vector and treat it as the quasi-stationary centre of gravity of that regime.

4.2 Two Crises in the Data

Table 2 reports the percentage change of each targeted moment from its baseline to its crisis window, for both episodes. Reading the two columns side by side motivates the entire estimation design.

The financial crisis is a textbook demand contraction, and a remarkably symmetric one. Unemployment rises by roughly three-fifths in *both* skill segments. Job-finding rates fall by about a quarter in both; separation rates rise by about a third in both; tightness halves in both. Plotting vacancies against unemployment, the 2008–10 observations lie close to the locus traced by the 2003–07 data: the economy slides *along* its Beveridge curve. Wages barely move. And the training share rises 13 percent—countercyclical enrolment, as the literature documents for every postwar recession in its sample windows.

COVID shares none of these signatures. Separation rates explode, and asymmetrically—+59 percent for unskilled workers, +75 percent for skilled ones. Job-finding rates do not fall at all. Tightness *rises* by a quarter to a third in both segments while unemployment is elevated: at the observed unemployment rates, vacancies sit far above what the pre-2020 Beveridge locus predicts, an outward shift that had not closed by the end of 2022. The skilled EE rate rises. And the training margin flips sign: the training share falls 4 percent, and the hazard of entering training from unskilled unemployment falls 17 percent.[4]

[4]In the FC pair the entry hazard is roughly flat (-5 percent) while the stock rises, because the unemployed

Table 2: Crisis signatures: percentage change of each moment from baseline to crisis window. CPS, ASEC, JOLTS, J2J; author's calculations. FC pair: 2003–07 → 2008–10. COVID pair: 2015–19 → 2020–22.

Moment	Δ% FC	Δ% COVID
Stocks		
Unemployment rate, total	+59	+35
Unemployment rate, unskilled	+61	+34
Unemployment rate, skilled	+60	+50
Skilled share	+6	+9
Training share	+13	−4
Wage variance, unskilled	−0	−1
Wage variance, skilled	+3	−4
Transitions		
Job-finding rate, unskilled	−27	+5
Separation rate, unskilled	+34	+59
Job-finding rate, skilled	−23	+1
Separation rate, skilled	+31	+75
EE rate, skilled	−23	+8
Training-entry hazard, unskilled	−5	−17
Wages		
Median wage, unskilled	+0	+3
Median wage, skilled	−1	−1
Log skill premium	−1	−4
Tightness		
θ_U	−51	+31
θ_S	−46	+26

[TODO: Insert Beveridge-by-skill figure (exists in descriptive notebook §8; export as plots/beveridge_by_skill.png and uncomment).]

One shock concept cannot rationalise both columns. A fall in aggregate demand moves an economy along its Beveridge curve: hiring slows, separations rise moderately, tightness falls. That is column one. Column two—separations spiking at *rising* tightness, job-finding flat, the curve itself displaced—requires the matching technology, the separation process, or both to have changed. The estimation below is built to let the data make this distinction parameter by parameter, and the model's two education margins are built to absorb its consequences: a demand shock works on the value of *entering* the skilled pipeline; a matching-technology shock works on the value of *remaining* in the skilled market.

4.3 Moments and Identification

Estimation targets the 18 moments of Table 2, computed within each window: seven stocks (three unemployment rates, the skilled share, the training share, and the two within-group wage variances), six transition rates (job-finding and separation by segment, the skilled EE rate, and the training-entry hazard), three wage moments (the two medians and the log skill premium), and the two tightness ratios.

The mapping from moments to parameters follows the model's structure. The skilled share and the training share discipline the training cutoff $\bar{x}$, and hence the interaction of the type distribution (a_ℓ, b_ℓ) with the cost level c; the training-entry hazard separates movements in c from composition shifts in the unemployed pool, because it conditions on the pool. Within each segment, the job-finding rate and tightness jointly identify the matching parameters (μ_j, η_j); the separation rate identifies the shock arrival rate λ_j together with the reservation rule it triggers; and the wage variances identify the match-quality distribution shapes $(\alpha_U; a_\Gamma, b_\Gamma)$ and the bargaining weights, since wages are linear in match quality given type. The skilled EE rate pins down the on-the-job search margin—the cost σ_S against the offer distribution. Median wages and the premium identify the productivity shifters P_U and γ_{P_S}. Tightness levels constrain vacancy costs through free entry. The non-employment payoffs (b_U, b_T, b_S) are identified from the level of reservation behaviour across segments: unemployment rates and the position of the lower tail of wages relative to the medians.

Two moments deserve comment. The training share is anchored to the NSC/IPEDS enrolment universe rather than taken raw from the CPS enrolment flags, which miss part-time

pool—the base to which the hazard applies—expands sharply. In the COVID pair both the stock and the hazard fall.

and non-degree enrolment; Appendix C describes the rescaling. The tightness pair requires allocating aggregate JOLTS vacancies across segments, which I do with the CPS industry-by-education composition; the same appendix reports the allocation matrix.

A testable implication. The classification below permits every matching, bargaining, and shock parameter to move between a baseline and its crisis window. A pure demand shock predicts that the matching block $(\mu_j, \eta_j, \beta_j, \sigma_S)$ should come back *unchanged* from the crisis re-estimation: a ride along a stable Beveridge curve. A reallocation shock predicts it should not. The estimation therefore does not assume what kind of crisis each episode was; it asks.

4.4 Parameter Classification and Multi-Regime Estimation

Parameters fall into four groups.

Table 3: Parameter classification

Group	Parameters	Treatment
Calibrated	r	4% annual; co SMM objectiv
Pre-estimated	ϕ	NSC/IPEDS to-degree: enr hazard across
	ν	CPS life-table age labour fo window.
Deep (fixed across each pair)	$a_\ell, b_\ell, b_U, b_T, b_S$	Estimated on fixed in the cr
Regime-specific	c; P_U, γ_{P_S}; k_U, k_S; λ_U, λ_S; α_U; a_Γ, b_Γ; μ_U, μ_S; η_U, η_S; β_U, β_S; σ_S	Re-estimated

The deep set is deliberately small: the population's ability distribution and the non-employment flow payoffs. These are the fundamentals I assume crises do not move. Everything else—technology, frictions, institutions of wage setting, the cost of education—is allowed to shift, because the premise of the paper is that crises may shift exactly such things. The discount rate cannot be separated from the flow payoffs by the targeted moments (their Jacobians are nearly collinear), so it is calibrated. The completion rate ϕ and turnover rate ν interact too tightly with the training cutoff and with r, respectively, to be identified from CPS moments alone, so both are pre-estimated from external data; Appendix C gives the recipes.

Estimation is by simulated method of moments,

$$\hat{\vartheta} = \arg\min_{\vartheta} \left[\hat{m} - m(\vartheta)\right]' W \left[\hat{m} - m(\vartheta)\right], \tag{27}$$

where $m(\vartheta)$ solves the stationary equilibrium at ϑ and W is the inverse of a scale-normalised, shrinkage-regularised estimate of the moment variance–covariance matrix, built from influence functions (Appendix C).

The procedure runs in two stages per crisis pair. Stage 1 estimates all deep and regime-specific parameters on the baseline window. Stage 2 fixes the deep parameters at their Stage 1 values and re-estimates the regime-specific block on the crisis window. The Stage 2 estimates are the paper's measurement of the crisis: which parameters moved, in which direction, by how much. Running the same two stages separately on the FC pair and the COVID pair, with deep parameters estimated independently on each baseline, yields the structural stability check: if the deep estimates from 2003–07 and 2015–19 agree, the classification is internally consistent across a twelve-year gap.

Given the two estimated steady states of a pair, the transition experiment of Section 3.7 is computed by switching the regime-specific block at T and tracing the economy's path from the baseline stationary distribution; the algorithm is in Appendix B.

5 Two Crises Through the Model

This section reads the two crises through the estimated model. I first describe the baseline equilibrium and the quality of the moment fit, then turn to the crisis re-estimations—which parameters moved in 2008 and in 2020—and finally to the computed transitions between the estimated steady states.[5]

5.1 Baseline Equilibrium

Figure 1 shows the endogenous segmentation of the economy at the estimated pre-FC parameters. The training cutoff splits the ability distribution into two pools: below $\bar{x}$, workers cycle between unskilled unemployment and unskilled jobs; above it, the population flows through the training pipeline into skilled unemployment and up the skilled job ladder. In the baseline the cross-market policy is dormant ($\underline{x} = 0$): no trained worker prefers unskilled

[5] Re-estimation of the current model version is in progress. Tables and figures will be replaced in place as estimates complete; the discussion in this section describes the configuration the descriptive evidence and the model's structure predict, and **[TODO:]**-flags mark every claim to be confirmed against the final estimates.

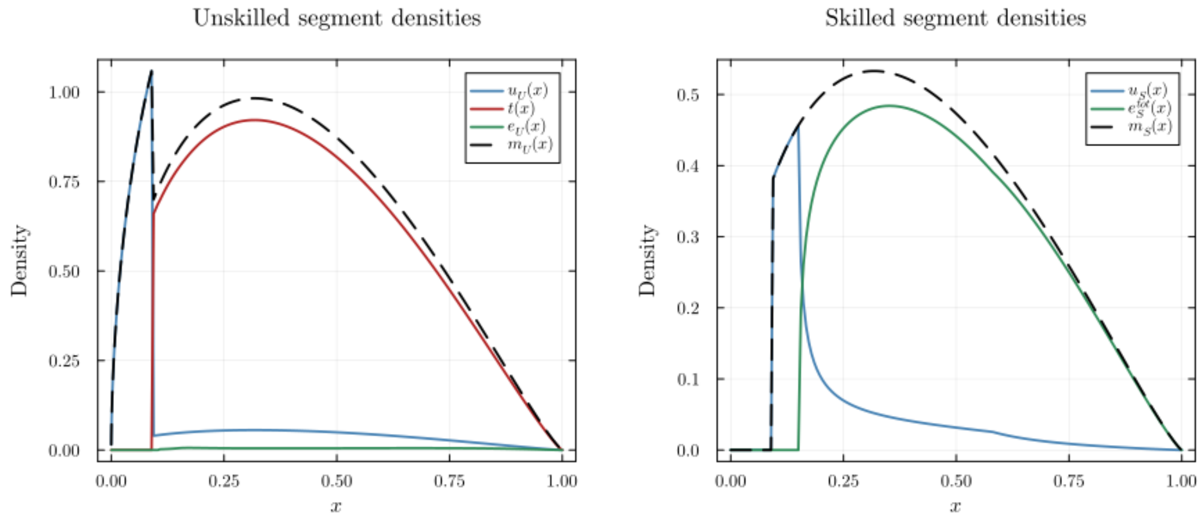

Figure 1: Cross-sectional densities over worker type x in the pre-FC baseline equilibrium: unskilled segment (left) and skilled segment (right). The training cutoff $\bar{x}$ partitions the two markets.

search, so the skilled segment retains everyone the pipeline delivers.

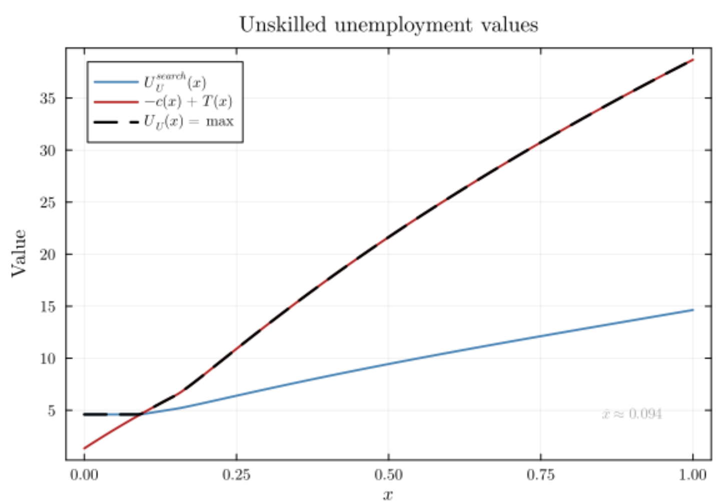

Figure 2: The training decision in the pre-FC baseline: value of unskilled search and net value of training $(-c(x) + T(x))$ over worker type. The curves cross at the training cutoff $\bar{x}$.

Figure 2 displays the decision that creates the cutoff. For low types the training cost is high and the skilled continuation value low, so unskilled search dominates; for high types the ordering reverses, and the gap widens with ability as the convex skilled technology pulls away from the linear unskilled one. The vertical distance between the curves above $\bar{x}$ measures how far from indifferent the inframarginal trainees are—a quantity that matters for the subsidy analysis of Section 6.

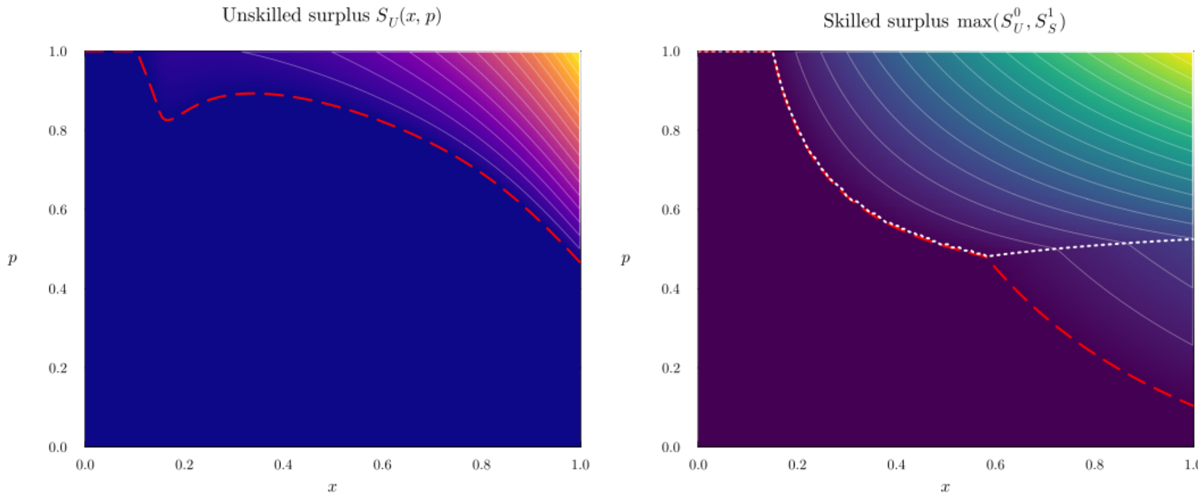

Figure 3: Match surplus over the (x, p) plane in the pre-FC baseline: unskilled surplus $S_U(x, p)$ (left) and skilled surplus $\max(S_S^0, S_S^1)$ (right). Overlaid curves mark the reservation quality $p_j^*(x)$ and, in the skilled panel, the on-the-job search cutoff $p_S^{oj}(x)$.

Figure 3 summarises both markets in one object each. The reservation curves slope downward: higher-ability workers tolerate worse matches. In the skilled panel the band between $p_S^*(x)$ and $p_S^{oj}(x)$ is the ladder-climbing region—employed, but still searching. How wide that band is, and how its width varies with x, governs the skilled EE rate the estimation targets.

5.2 Moment Fit

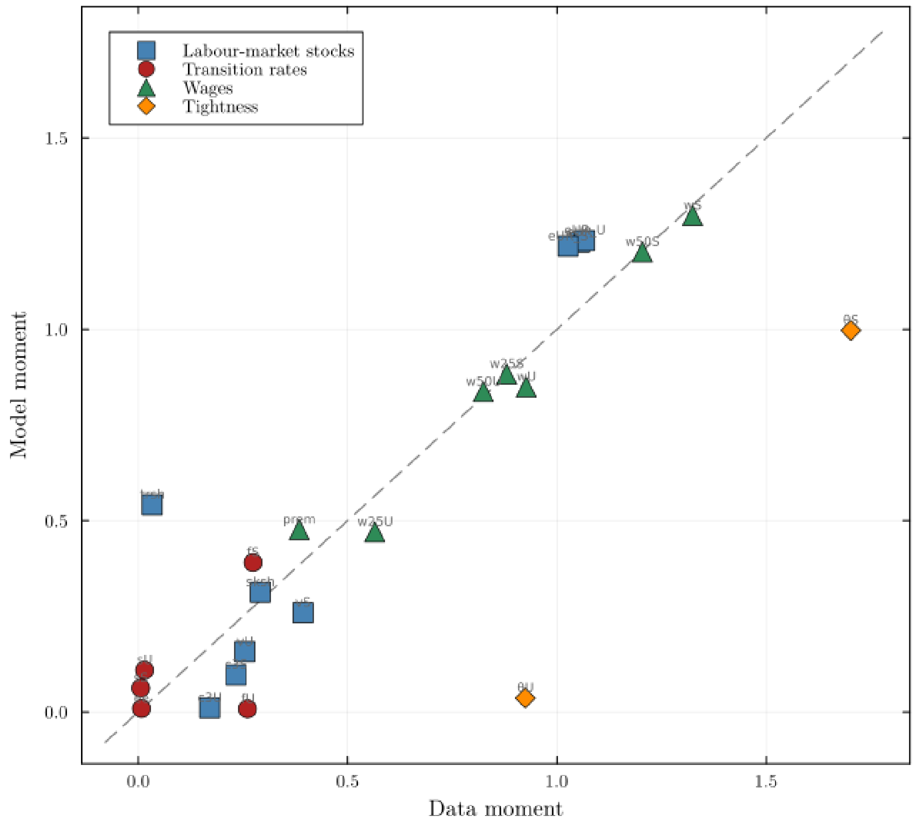

Figure 4: Model fit: data moments versus model counterparts for the pre-financial-crisis baseline. Points on the 45-degree line are matched exactly.

Tables 4 and 5 report the fit for the two baseline windows side by side. The comparison is the first half of the structural stability check: the deep parameters are estimated separately on each baseline, so similar fit quality across the two expansions—twelve years apart, with a markedly tighter labour market in the second—indicates that the deep/regime split is not absorbing secular change into the wrong block.

[TODO: Once the new estimates are in: brief block-by-block reading of the fit (stocks, transitions, wages, tightness), noting which blocks bind. Keep it to two paragraphs; the table carries the detail.]

Table 4: Model Fit (A): Labour-Market Stocks and Transition Rates

Moment	Financial Crisis			COVID		
	Data	Model	Diff.	Data	Model	Diff.
Labour-market stocks						
exp(UR total)	1.0543	1.2274	+0.1731	1.0462	1.3041	+0.2579
	(0.0057)	(0.0025)		(0.0066)	(0.0019)	
exp(UR unskilled)	1.0661	1.2321	+0.1661	1.0582	1.3648	+0.3066
	(0.0063)	(0.0025)		(0.0086)	(0.0026)	
exp(UR skilled)	1.0264	1.2171	+0.1907	1.0250	1.1496	+0.1246
	(0.0043)	(0.0026)		(0.0027)	(0.0013)	
Skilled share	0.2915	0.3133	+0.0218	0.3596	0.2652	-0.0945
	(0.0208)	(0.0011)		(0.0295)	(0.0007)	
Training share	0.0334	0.5416	+0.5081	0.0450	0.4584	+0.4134
	(0.1061)	(0.0018)		(0.0903)	(0.0011)	
Var(w) unskilled	0.2543	0.1578	-0.0965	0.2860	0.1555	-0.1305
	(0.0643)	(0.0259)		(0.0432)	(0.0371)	
CM3(w) unskilled	0.1709	0.0110	-0.1599	0.2624	0.0402	-0.2222
	(0.0740)	(0.0124)		(0.0454)	(0.0394)	
Var(w) skilled	0.3934	0.2611	-0.1323	0.4498	0.3023	-0.1475
	(0.0477)	(0.0315)		(0.0480)	(0.0402)	
CM3(w) skilled	0.2337	0.0980	-0.1357	0.3514	0.1400	-0.2114
	(0.0470)	(0.0359)		(0.0320)	(0.0272)	
Transition rates						
JFR unskilled	0.2610	0.0083	-0.2528	0.2565	0.0639	-0.1926
	(0.0849)	(0.0177)		(0.0846)	(0.0099)	
Sep. rate unskilled	0.0151	0.1097	+0.0945	0.0132	0.0241	+0.0109
	(0.0634)	(0.0384)		(0.1165)	(0.0155)	
JFR skilled	0.2741	0.3907	+0.1166	0.2717	0.4577	+0.1860
	(0.1500)	(0.0704)		(0.1398)	(0.0956)	
Sep. rate skilled	0.0062	0.0628	+0.0566	0.0059	0.0428	+0.0369
	(0.1246)	(0.0727)		(0.1617)	(0.0326)	
EE rate skilled	0.0080	0.0052	-0.0027	0.0085	0.0032	-0.0053
	(0.0624)	(0.0624)		(0.0366)	(0.0366)	

Table 5: Model Fit (B): Wages and Tightness

Moment	Financial Crisis			COVID		
	Data	Model	Diff.	Data	Model	Diff.
Wages						
Mean wage unskilled	0.9265	0.8493	-0.0772	0.9166	0.8030	-0.1136
	(0.0360)	(0.0178)		(0.0315)	(0.0269)	
Mean wage skilled	1.3239	1.2951	-0.0289	1.3144	1.1705	-0.1440
	(0.0346)	(0.0208)		(0.0289)	(0.0258)	
p25 wage unskilled	0.5652	0.4709	-0.0943	0.5482	0.4779	-0.0703
	(0.0393)	(0.0190)		(0.0453)	(0.0392)	
p25 wage skilled	0.8800	0.8776	-0.0024	0.8411	0.7201	-0.1210
	(0.0392)	(0.0231)		(0.0278)	(0.0264)	
p50 wage unskilled	0.8244	0.8371	+0.0127	0.7920	0.7337	-0.0583
	(0.0384)	(0.0194)		(0.0338)	(0.0285)	
p50 wage skilled	1.2042	1.2008	-0.0033	1.1594	1.0480	-0.1114
	(0.0377)	(0.0237)		(0.0298)	(0.0257)	
Skill premium	0.3846	0.4743	+0.0897	0.3851	0.3959	+0.0108
	(0.0136)	(0.0113)		(0.0172)	(0.0141)	
Tightness						
θ_U	0.9244	0.0367	-0.8878	1.5849	5.6596	+4.0747
	(0.2101)	(0.0880)		(0.2524)	(0.0567)	
θ_S	1.7020	0.9972	-0.7048	2.8943	1.3089	-1.5854
	(0.2508)	(0.1374)		(0.1508)	(0.0555)	

Table 6: Parameter Estimates (A): Calibrated and Deep Structural

Parameter	Description	Financial Crisis			COVID		
		Baseline	Crisis	Δ	Baseline	Crisis	Δ
Externally calibrated							
r	Discount rate	0.00417	0.00417	—	0.00417	0.00417	—
ν	Demographic exit rate	0.03841	0.03841	—	0.03841	0.03841	—
φ	Training completion rate	0.02222	0.02222	—	0.02222	0.02222	—
Deep structural							
a_ℓ	Worker type shape	1.50538	1.50538	—	1.06876	1.50538	—
b_ℓ	Worker type shape	2.87219	2.87219	—	2.56120	2.87219	—
c	Training cost coeff.	2.94357	2.94357	—	2.91536	2.94357	—
b_U	Unskilled UI flow	0.19600	0.19600	—	0.23613	0.19600	—
b_T	Training flow	1.02176	1.02176	—	1.11815	1.02176	—
b_S	Skilled UI flow	0.56562	0.56562	—	0.43460	0.56562	—
μ_U	Unskilled matching eff.	0.04330	0.04330	—	0.02687	0.04330	—
η_U	Unskilled matching elas.	0.50000	0.50000	—	0.50000	0.50000	—
β_U	Unskilled bargaining	0.50000	0.50000	—	0.50000	0.50000	—
μ_S	Skilled matching eff.	1.18859	1.18859	—	0.84933	1.18859	—
η_S	Skilled matching elas.	0.50000	0.50000	—	0.50000	0.50000	—
β_S	Skilled bargaining	0.50000	0.50000	—	0.50000	0.50000	—
σ_S	OJS flow cost	1.10000	1.10000	—	1.10000	1.10000	—

Table 7: Parameter Estimates (B): Regime-Specific

Parameter	Description	Financial Crisis			COVID		
		Baseline	Crisis	Δ	Baseline	Crisis	Δ
Regime-specific							
P_U	Unskilled productivity	2.04377	1.90910	-0.13466	2.73686	2.09363	-0.64323
P_S	Skilled productivity	3.75855	4.05600	+0.29746	3.95147	4.95622	+1.00476
α_U	Unskilled damage shape	5.89456	5.94021	+0.04565	5.43224	7.78774	+2.35551
a_Γ	Skilled offer shape	2.75946	2.91353	+0.15407	3.39435	2.14112	-1.25323
b_Γ	Skilled offer shape	0.70915	1.14897	+0.43981	1.00198	0.82445	-0.17753
k_U	Unskilled vacancy cost	0.00979	0.00351	-0.00628	0.00207	0.00100	-0.00107
λ_U	Unskilled damage rate	0.29197	0.18142	-0.11054	0.32073	0.92715	+0.60642
k_S	Skilled vacancy cost	0.17944	0.19874	+0.01930	0.16986	0.92440	+0.75454
ξ_S	Skilled exog. sep. rate	0.00000	0.00000	+0.00000	0.00000	0.00000	-0.00000
λ_S	Skilled quality shock	0.42931	0.38727	-0.04204	0.43895	0.49556	+0.05661

5.3 Parameter Estimates: What Moved in Each Crisis

Tables 6 and 7 report the estimates for all four windows; the crisis columns hold the deep block fixed and re-estimate the rest. The reading the descriptive signatures of Section 4.2 predict is as follows.

The financial crisis: a demand shock. The FC column should be carried by the aggregate-state block—a fall in P_U, movements in the vacancy costs k_j, and a moderate rise in the destruction margin—while the matching technology (μ_j, η_j), the bargaining weights, and the OJS cost return close to their baseline values. That configuration is what "a ride along the Beveridge curve" looks like in parameter space: contact rates fall because tightness falls, not because matching deteriorated. With the skilled option value broadly intact, the cross-market margin stays dormant, and the crisis operates on the education margin only through the entry channel: the value of unskilled search falls relative to the training pipeline, $\bar{x}$ falls, and the training share rises. **[TODO: Confirm against final FC estimates; report the handful of parameters whose movement is economically large, with one sentence each on identification.]**

COVID: a matching-efficiency and reallocation shock. The COVID column should look different in kind, not in degree: a fall in the matching efficiencies μ_j (the outward Beveridge shift proper), a sharp rise in λ_S (the separation spike), and a deterioration of the skilled offer distribution (a_Γ, b_Γ), with productivity shifts playing a secondary role. This is the configuration that compresses the skilled tail integral $\mathcal{I}_S(x)$ for low-ability trained workers while leaving the unskilled contact rate intact—precisely the geometry that activates the cross-market choice. **[TODO: Confirm which subset of $(\mu_j, \eta_j, \beta_j, \sigma_S, \lambda_j, \alpha_U, a_\Gamma, b_\Gamma)$ moves; the testable implication of Section 4.3 is scored here.]**

5.4 The Two Margins Across Regimes

Table 8: Training Cutoff $\bar{x}$

	Baseline	Crisis	Δ
Financial Crisis	0.0921	0.0921	+0.0000
COVID	0.0967	0.0922	-0.0046

The model condenses each crisis into the movement of two thresholds. The financial crisis moves the *entry* margin: $\bar{x}$ falls, more workers train, and the training share rises along the transition as the pipeline fills. The cross-market margin stays at $\underline{x} = 0$. COVID moves the *exit* margin: the compressed skilled option value pushes $\underline{x}$ above zero, trained workers below it abandon skilled search, and the training share falls—even though the entry cutoff $\bar{x}$ need not rise. The model thus reproduces the sign flip in Table 2 with one mechanism per crisis, both estimated, neither imposed. **[TODO: Insert $\bar{x}$ (and $\underline{x}$ for COVID) values across the four windows once computed; one sentence comparing the magnitude of the $\bar{x}$ shift to the training-share movement it implies.]**

This reading also explains why the entry *hazard* fell slightly in the FC crisis window while the stock rose (Table 2): the newly unemployed in 2008–10 were drawn disproportionately from low-x types for whom training is never optimal, diluting the hazard even as the cutoff fell. The hazard conditions on a pool whose composition the crisis itself shifts—one reason the estimation targets the hazard and the stock jointly.

5.5 Transition Dynamics

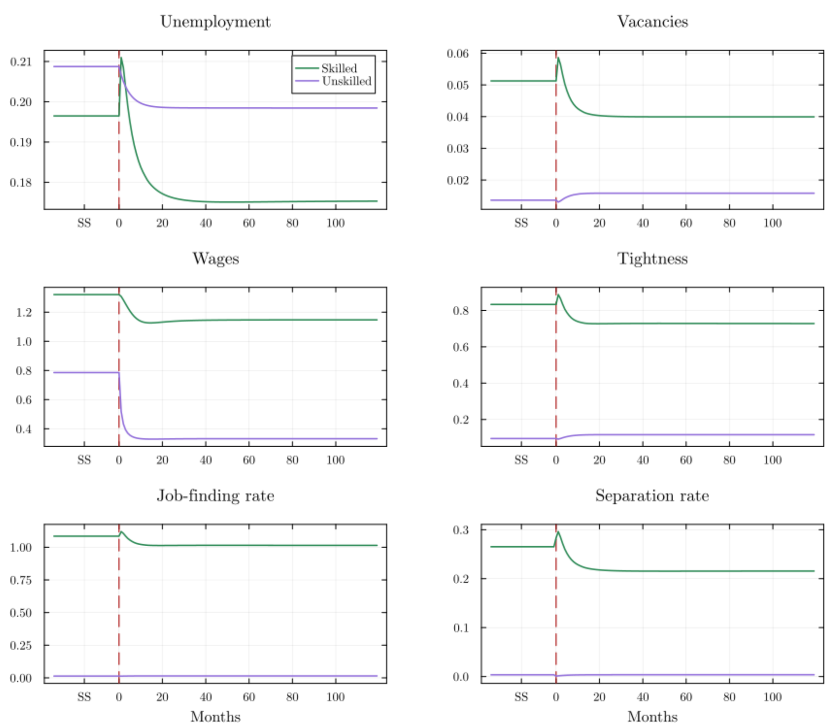

Figure 5: Transition dynamics after the financial-crisis regime switch: model-implied paths of unemployment, vacancies, wages, tightness, job-finding, and separation rates, with pre- and post-switch steady states marked.

Figure 5 traces the economy's adjustment after the unexpected switch to the FC regime. The mechanics follow Section 3.7. At the instant of the switch, value functions and the four policy rules jump; the distributions are inherited. The separation rate spikes on impact:

the employed stock carries the baseline distribution of match qualities, and the new, higher reservation thresholds render its lower tail unprofitable—a stock-revaluation effect that dissipates as destroyed matches are replaced by hires formed under crisis conditions. Tightness drops through free entry and then recovers partially as the composition of the unemployed pool improves. Unemployment rises fast and converges slowly. The skilled segment lags the unskilled one: its inflow responds to the lower $\bar{x}$ only at the pipeline rate $\phi+\nu$, so the skilled share drifts upward for years after the switch—the model's account of crisis-driven hysteresis in skill composition.

[TODO: Add the COVID transition figure (proposed name plots/model_decomposition_covid_ once the COVID pair is estimated, and uncomment.]

The COVID transition differs in one structural respect: the drain term $d(x,t)f_U(t)u_S(x,t)$ in the law of motion (26) is active. On the affected types, the skilled segment empties at the unskilled job-finding rate rather than the demographic rate—an order-of-magnitude difference in speed. The model therefore predicts that the skilled share's response to COVID is front-loaded, while its response to the financial crisis was back-loaded; the half-life of the trained stock is no longer a single number but a margin-dependent object. **[TODO: Quantify both half-lives from the computed transitions; compare with the decay rates estimated directly from the data in the descriptive analysis.]**

6 Policy

The estimated model prices two families of policy instruments. Non-employment payoffs (b_U, b_S) enter every reservation decision and both education margins; education subsidies (b_T, c) target the training pipeline directly. This section works out what the model says about each—analytically where the structure permits, quantitatively where it does not. Throughout, the experiments perturb one estimated baseline and compare stationary equilibria; the transition costs of moving between them are computable with the machinery of Section 3.7 but are not the focus here.

6.1 Outside Options, Unemployment Duration, and Match Quality

Whether unemployment insurance subsidises productive selectivity or unproductive waiting is a long-standing question. The sharpest evidence is quasi-experimental: Nekoei and Weber (2017) find that a nine-week extension of Austrian UI raises reemployment wages by about half a percent, and read the result as the net of two offsetting forces—workers hold out for better matches, but longer spells erode prospects. Schmieder et al. (2016) find the opposite net sign in German data. A structural account of *both* forces, and of why their balance can differ across settings, requires exactly the objects this model estimates: reservation rules, offer distributions, and free-entry responses. This subsection derives the relevant formulas and then turns the estimated model loose on them.

The reservation response in closed form

Hold tightness fixed for the moment (the partial-equilibrium experiment) and consider the skilled market; the unskilled case is symmetric. Raising b_S raises the value of skilled unemployment, which raises the reservation quality $p_S^*(x)$. Differentiating the surplus recursion (11) together with the unemployment value (18) yields a closed form (Appendix A):

Proposition 3 (Reservation response to outside options)**.** *Fix tightness, and let* $\delta_S(x) \equiv \lambda_S \Gamma(p_S^*(x))$ *denote the endogenous destruction hazard and* $q_S^a(x) \equiv 1 - \Gamma(p_S^*(x))$ *the acceptance probability. On the* $d = 0$*, no-OJS region,*

$$\frac{\partial p_S^*(x)}{\partial b_S} \;=\; \frac{r + \nu + \lambda_S}{\big[\, r + \nu + \delta_S(x) + \beta_S\, \kappa_S\, q_S^a(x) \,\big]\; \gamma_{P_S}\, x^{\gamma_{P_S}}} \;>\; 0, \tag{28}$$

and the analogous expression holds in the unskilled market with $(\lambda_U, G, f_U, \beta_U, P_U\, x)$ *in place of* $(\lambda_S, \Gamma, \kappa_S, \beta_S, \gamma_{P_S} x^{\gamma_{P_S}})$*. The response is strictly decreasing in* x*.*

Two features of (28) organise everything that follows. First, the denominator contains the worker's share of the market option value, $\beta_S \kappa_S q_S^a$: workers who expect offers to keep arriving let the market, not the benefit, set their standards, so outside options move reservation behaviour least where contact rates are high. Second, the production term $\gamma_{P_S} x^{\gamma_{P_S}}$ makes the response fall with ability. A flat increase in b_S is a large change relative to a low-x worker's flow output and a rounding error relative to a high-x worker's. Outside options are reservation-relevant for exactly the workers the skilled market values least—the same workers, Proposition 2 showed, who sit nearest the cross-market exit.

Duration and match quality

The expected unemployment duration and the expected quality of the match that ends it follow from the reservation rule. A skilled unemployed worker of type x (with $d(x) = 0$)

exits to employment at hazard $\kappa_S\, q_S^a(x)$, so

$$D_S(x) = \frac{1}{\kappa_S\left(1-\Gamma(p_S^*(x))\right)}, \qquad d\ln D_S(x) = \underbrace{-\,d\ln\kappa_S}_{\text{market channel}} + \underbrace{\frac{\gamma\big(p_S^*(x)\big)}{1-\Gamma\big(p_S^*(x)\big)}\, dp_S^*(x)}_{\text{selectivity channel}}, \tag{29}$$

and the expected quality of the accepted match is the truncated mean

$$Q_S(x) = \mathbb{E}_\Gamma\big[\,p \mid p \geq p_S^*(x)\,\big], \qquad \frac{\partial Q_S(x)}{\partial p_S^*} = \frac{\gamma(p_S^*)}{1-\Gamma(p_S^*)}\big(Q_S(x) - p_S^*(x)\big) > 0. \tag{30}$$

Equation (29) is the Nekoei–Weber decomposition as the model writes it. The selectivity channel is Proposition 3 composed with the offer distribution's hazard; it lengthens spells and improves accepted matches in the same proportion-generating act. The market channel is the general-equilibrium response of κ_S through free entry: a higher b_S compresses surpluses, vacancy posting falls, and every skilled searcher—selective or not—waits longer. Which channel dominates is a quantitative question, settled by the estimated offer distribution's hazard at the relevant cutoffs and the free-entry elasticity.

The unskilled market splits the same forces across different margins. Entry quality is fixed at the frontier, so b_U does not act on an acceptance margin: spell duration $D_U = 1/f_U$ moves *only* through the market channel. The selectivity channel operates instead at separation: a higher b_U raises $p_U^*(x)$, so marginal matches dissolve sooner and the surviving stock of matches is better on average. Outside options buy match quality in both markets—through pickier acceptance in one, through quicker abandonment of damaged matches in the other—and buy longer non-employment in both.

Steering the education margins

The cross-market structure adds a result that single-market treatments of UI generosity cannot express: the two payoffs steer the education margins in opposite directions.

Proposition 4 (Margin steering)**.** *Fix tightness. Then: (i) $\partial\bar{x}/\partial b_U > 0$ and $\partial\underline{x}/\partial b_U > 0$: a higher unskilled payoff discourages entry into training and accelerates exit from the skilled market. (ii) $\partial\bar{x}/\partial b_S < 0$; and $\partial\underline{x}/\partial b_S < 0$ provided $f_U\left(r+\nu+\delta_S(\underline{x})\right) > \beta_S\,\kappa_S\,q_S^a(\underline{x})\,(r+\nu)$, a condition satisfied whenever the unskilled contact rate is not small relative to the worker's share of the skilled acceptance hazard: a higher skilled payoff encourages entry and retains marginal trained workers.*

The asymmetry has a simple source. b_U enters the training comparison only through the value of unskilled search, and enters the cross-market comparison only through the value of the unskilled job a defecting worker would take—both times on the unskilled side of the ledger. b_S enters through the training value $T(x)$ and through the skilled unemployment value a worker keeps by staying—both times on the skilled side. The flow payoff b_S is paid in skilled unemployment regardless of which market the worker searches, but a worker who searches unskilled expects to leave unemployment sooner, so she discounts the b_S stream more heavily; that is why a higher b_S tilts her toward staying (Appendix A).

For crisis policy the proposition reads as an instrument-assignment rule. A demand crisis attacks the entry margin: the value of unskilled search collapses, $\bar{x}$ falls, and the safety-net question is the classic duration-versus-quality trade-off of equation (29), for which b_U is the relevant dial. A reallocation crisis attacks the retention margin: $\mathcal{I}_S(x)$ compresses, $\underline{x}$ rises, and marginal trained workers shed their training—an outcome b_U generosity *accelerates* and b_S generosity retards. The model prices the choice between cushioning the unskilled and preserving the trained, and the price depends on which crisis the economy is in.

Quantitative exercise

I solve the model on a grid of (b_U, b_S) pairs around each estimated baseline and report, for each pair, expected durations by market, accepted match quality, the two cutoffs, tightness, unemployment, and welfare. **[TODO: Exercise design to be finalised together with the figure choice (see suggestions report): leading candidate is a two-panel figure—(a) the duration–quality menu traced by varying each payoff, plotted as a curve in (D, Q) space per market; (b) iso-welfare (or iso-output) contours in the (b_U, b_S) plane computed separately under FC-regime and COVID-regime parameters, whose differing gradients display the instrument-assignment result of Proposition 4.]**

6.2 Education Subsidies: Stipends versus Cost Reduction

Search frictions hold up ex-ante investments: when prices are set by ex-post bargaining, the investing side cannot appropriate the full return (Acemoglu and Shimer, 1999), and in education-in-search settings the laissez-faire level of schooling is generically inefficient, with a direct education subsidy the natural corrective (Charlot and Malherbet, 2013). The model prices two subsidy designs that policy debates treat as interchangeable: a flow stipend paid while training (b_T) and an up-front reduction in the training cost (the parameter c in $c(x) = (1 - x)e^{c-x}$). Both move only the training comparison $-c(x) + T(x) \gtrless U_U^{\text{search}}(x)$; neither touches market structure directly. They are nonetheless not equivalent, and the model says exactly how they differ.

Proposition 5 (Marginal equivalence and fiscal cost)**.** *(i) A small stipend increase* db_T *moves the training cutoff exactly as much as an up-front cost reduction worth* $db_T/(r+\phi+\nu)$ *at the margin* $\bar{x}$*. (ii) Let* $\bar{c}_e \equiv \int_{\bar{x}}^{1} c(x)\,\omega(x)\,dx$ *be the entry-weighted average training cost, with weights* $\omega(x) \propto \tau(x)u_U(x)$*. In stationary equilibrium, the ratio of the stipend's fiscal flow cost to that of the cutoff-equivalent proportional cost subsidy is*

$$\frac{\text{stipend cost}}{\text{cost-subsidy cost}} \;=\; \frac{r+\phi+\nu}{\phi+\nu} \;\cdot\; \frac{c(\bar{x})}{\bar{c}_e} \;>\; 1. \tag{31}$$

Part (i) is the present-value logic of equation (4): a trainee receives the stipend until completion or exit, events with combined hazard $\phi + \nu$, discounted at r. Part (ii) decomposes the stipend's disadvantage into two factors, each with its own economics. The first factor, $(r + \phi + \nu)/(\phi + \nu)$, is a discounting wedge: the government pays the stipend as an undiscounted flow over spells of expected length $1/(\phi + \nu)$, while the workers it is meant to move value that flow at the discounted rate. The second factor, $c(\bar{x})/\bar{c}_e > 1$, is targeting: training costs decline in ability, so the marginal trainee—the one at $\bar{x}$, the only one whose behaviour the subsidy changes—has the *highest* cost among all entrants. A proportional cost subsidy automatically concentrates its outlay on her; a stipend pays every trainee the same flow, spending most of its budget on inframarginal high-ability entrants who would have trained at the old terms. Proofs are in Appendix A.

The proposition ranks the instruments per unit of cutoff movement at the margin; it is not yet a welfare ranking, for two reasons the quantitative exercise addresses. First, large subsidies move $\bar{x}$ discretely, and the two instruments then recruit different inframarginal populations with different spell lengths. Second, both instruments share a general-equilibrium fate: the additional trainees arrive in skilled unemployment, and whether the skilled contact rate holds up depends on the vacancy-creation response in (17). If entry is sluggish, congestion claws back part of any subsidy, a channel competitive treatments of education policy do not price (Heckman et al., 1998; Abbott et al., 2019). The medium-run profile of programme effects documented by Card et al. (2018)—small on impact, larger two to three years out—is what this structure predicts: subsidised entrants start at the bottom of the skilled ladder and their wage gains accrue as they climb. And the timing evidence of Lindner and Reizer (2020)—that frontloading a fixed UI budget shortens non-employment without degrading match quality—is the reduced-form counterpart of the discounting wedge in (31): recipients act on the present value, the budget pays the flow.

Quantitative exercise

For each instrument I solve stationary equilibria at matched fiscal budgets around the estimated baseline and report the cutoff, the training stock, the skilled share, tightness and unemployment in both markets, wages, and welfare by ability group.

Table 9: Education-subsidy counterfactuals — Baseline-FC

	Baseline	+5% bT	+10% bT	+20% bT	-5% c(x)	-10% c(x)	-20% c(x)
$\bar{x}$	0.1031	0.0703	0.0517	0.0054	0.0703	0.0434	0.0002
ur	0.2049	0.1771	0.1633	0.1403	0.1771	0.1577	0.1397
Skilled share	0.3133	0.3303	0.3388	0.3525	0.3301	0.3416	0.3524
Training share	0.5416	0.5710	0.5856	0.6094	0.5705	0.5905	0.6091
θ_U	0.0367	0.0281	0.0150	0.0073	0.0541	0.0833	0.1608
θ_S	0.9972	0.7594	0.6794	0.5810	0.7572	0.6520	0.5764
$\bar{w}_U$	0.8366	1.0690	1.2759	1.4635	1.0268	1.1503	1.2526
$\bar{w}_S$	1.2824	1.2735	1.2695	1.2649	1.2730	1.2680	1.2637
f_U	0.0083	0.0073	0.0053	0.0037	0.0101	0.0125	0.0174
f_S	0.3907	0.2783	0.2429	0.1988	0.2782	0.2302	0.1973
δ_U	0.1097	0.1718	0.1832	0.2074	0.1355	0.1419	0.1307
δ_S	0.0628	0.0547	0.0512	0.0470	0.0544	0.0504	0.0470

Notes: Policy A raises b_T by the stated percentage. Policy B reduces e^c (and hence $c(x) = (1-x)e^{c-x}$) by the stated percentage. † indicates non-convergence.

[TODO: Refresh table with re-estimated baseline; add the matched-budget comparison and, if the suggestions-report figure is adopted, the per-dollar cutoff-shift figure.]

7 Conclusion

This paper treats crises as what they look like in labour-market data: unanticipated, persistent shifts in structural parameters, not realisations of a business cycle. It builds a search model in which the boundary between an unskilled and a skilled market moves in both directions—into skill through a training cutoff, out of it through a cross-market search choice that makes training obsolete—and estimates the model as a pair of steady states around each of two crises, holding a small set of deep parameters fixed and letting the data decide which technology, friction, and institutional parameters moved. The two crises receive different answers, and the difference explains a fact that the countercyclical-enrolment literature does not cover: training rose in 2008 and fell in 2020. A demand shock works on the entry margin; a matching-efficiency shock works on the retention margin, and the computed transitions show the second margin draining the skilled segment at the unskilled job-finding rate—far faster than demographic turnover, which is the only clock a one-margin model has.

The estimated structure pays for itself in the policy analysis. The reservation response to outside options has a closed form that falls with ability, so non-employment payoffs bind hardest exactly where the cross-market exit looms; the duration–quality trade-off decomposes into a selectivity and a market channel, giving structural content to the quasi-experimental UI literature; and the two payoffs steer the two education margins in opposite directions, so the choice between cushioning unskilled workers and preserving trained ones is a real choice whose terms depend on the crisis. On the education-subsidy side, a stipend and a cost reduction that are equivalent at the margin differ in fiscal cost by a closed-form factor—a discounting wedge times a targeting wedge—with general-equilibrium congestion in the skilled market then determining how much of either instrument survives.

The model abstracts from margins that would enrich it. Ability is fixed: a skill multiplier evolving with employment experience would let displacement scar workers as well as reallocate them, connecting the framework to the displacement-cost literature. Firms are homogeneous within markets: firm heterogeneity would let the model speak to sorting and to which employers absorb the cross-market movers. Regime switches are permanent and unanticipated: temporary or partially anticipated switches would sharpen the welfare analysis of crisis-contingent policy. These are extensions of the machinery, not repairs to it. The broader point stands without them: models that treat a crisis as a parameter shift, and estimate which parameters shifted, can say when different labour-market policies work and why—a question that steady states and business cycles, by construction, cannot pose.

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A Derivations and Proofs

This appendix collects the derivations behind the equations of Section 3 and the proofs of Propositions 1–5. Throughout, $\delta_U(x) \equiv \lambda_U G(p_U^*(x))$ and $\delta_S(x) \equiv \lambda_S \Gamma(p_S^*(x))$ denote the endogenous destruction hazards, and $q^a_S(x) \equiv 1-\Gamma(p^*_S(x))$ the skilled acceptance probability. In the unskilled market every contact is accepted, $q^a_U \equiv 1$, because matches form at the frontier.

A.1 Wage-free surplus recursions and wage schedules

Unskilled market. Worker and firm values of a match at quality p satisfy

$$(r+\nu+\lambda_U)\, E_U(x,p) = w_U(x,p) + \lambda_U \left[U_U(x)\, G(p^*_U(x)) + \int_{p^*_U(x)}^{1} E_U(x,p')\, dG(p') \right], \tag{32}$$

$$(r+\nu+\lambda_U)\, J_U(x,p) = P_U x p - w_U(x,p) + \lambda_U \int_{p^*_U(x)}^{1} J_U(x,p')\, dG(p'). \tag{33}$$

Define $S_U = (E_U - U_U) + J_U$. Adding (32) and (33) and substituting $E_U + J_U = S_U + U_U$ gives the wage-free recursion (7) of the text. Nash bargaining imposes $\beta_U J_U = (1-\beta_U)(E_U - U_U)$, i.e. $E_U - U_U = \beta_U S_U$ and $J_U = (1-\beta_U) S_U$. Multiplying (7) by $(1-\beta_U)$ and comparing with (33) recovers the wage schedule (9); linearity in p follows because the only p-dependence is through flow output. Setting $S_U(x, p^*_U) = 0$ in (7) gives the reservation rule (8).

Unskilled search value. Rearranging (6) and substituting $E_U(x,1) = U_U(x) + \beta_U S_U(x,1)$,

$$(r+\nu+f_U)\, U^{\text{search}}_U(x) = b_U + f_U \big(U_U(x) + \beta_U S_U(x,1) \big). \tag{34}$$

On the search region $x \le \bar{x}$, where $U_U = U^{\text{search}}_U$, this reduces to

$$(r+\nu)\, U^{\text{search}}_U(x) = b_U + \beta_U f_U\, S_U(x,1). \tag{35}$$

Skilled market. Under search regime $s \in \{0,1\}$, with $\pi(x) \equiv \gamma_{P_S} x^{\gamma_{P_S}}$, the worker and firm values satisfy

$$(r+\nu+\lambda_S)\, E_S^0(x,p) = w_S^0(x,p) + \lambda_S \left[U_S(x)\, \Gamma(p_S^*(x)) + \int_{p_S^*(x)}^1 E_S(x,p')\, d\Gamma(p') \right], \tag{36}$$

$$\begin{aligned}(r+\nu+\lambda_S)\, E_S^1(x,p) = w_S^1(x,p) - \sigma_S + \lambda_S \left[U_S(x)\, \Gamma(p_S^*(x)) + \int_{p_S^*(x)}^1 E_S(x,p')\, d\Gamma(p') \right] \\ + \kappa_S \int_{\max\{p,\, p_S^*(x)\}}^1 \big(E_S(x,\tilde p) - E_S^1(x,p) \big)\, d\Gamma(\tilde p),\end{aligned} \tag{37}$$

$$(r+\nu+\lambda_S)\, J_S^0(x,p) = \pi(x)\, p - w_S^0(x,p) + \lambda_S \int_{p_S^*(x)}^1 J_S(x,p')\, d\Gamma(p'), \tag{38}$$

$$\big(r+\nu+\lambda_S+\kappa_S(1-\Gamma(p))\big) J_S^1(x,p) = \pi(x)\, p - w_S^1(x,p) + \lambda_S \int_{p_S^*(x)}^1 J_S(x,p')\, d\Gamma(p'), \tag{39}$$

with $E_S = \max\{E_S^0, E_S^1\}$ and $J_S = J_S^{s^*}$. Summing each worker–firm pair, subtracting the unemployment value, and using the Nash split $E_S^s - U_S = \beta_S S_S^s$, $J_S^s = (1-\beta_S) S_S^s$ yields (11)–(12). The wage schedules (14)–(15) follow by the same comparison used in the unskilled market. The OJS wage discount,

$$w_S^1(x,p) - w_S^0(x,p) = (1-\beta_S)\,\sigma_S - \beta_S(1-\beta_S)\,\kappa_S \int_p^1 S_S(x,p')\, d\Gamma(p') \;\le\; 0 \quad \text{on } p < p_S^{oj}(x), \tag{40}$$

is non-positive on the search region because optimality of search there requires $\beta_S \kappa_S \int_p^1 S_S\, d\Gamma \geq \sigma_S$.

The two unemployment branches. Conditional on the search-market policy, the skilled unemployment Bellman is

$$(r+\nu)\, U_S(x) = b_S + (1-d(x))\, \kappa_S \int_{p_S^*(x)}^1 \beta_S S_S(x,\tilde p)\, d\Gamma(\tilde p) + d(x)\, f_U\, \big(E_U(x,1) - U_S(x)\big).$$

Solving each branch for U_S gives (18)–(19); the $d=1$ branch embeds acceptance of every unskilled offer, which is optimal for any worker who prefers $d=1$ (a worker who would reject would deviate to $d=0$).

A.2 Stationary flow balance

For each x, with $\tau(x) = \mathbf{1}\{x > \bar{x}\}$ and $d(x) = \mathbf{1}\{x \le \underline{x}\}$:

Training:
$$0 = \tau(x)\, u_U(x) - (\phi + \nu)\, t(x), \tag{41}$$

Untrained unemployment:
$$\big(f_U + \tau(x) + \nu\big)\, u_U(x) = \nu\, \ell(x) + \delta_U(x)\, e_U(x), \tag{42}$$

Skilled unemployment:
$$\Big(\nu + (1-d)\, \kappa_S q_S^a(x) + d\, f_U\Big) u_S(x) = \phi\, t(x) + \delta_S(x)\, e_S^{\mathrm{tot}}(x), \tag{43}$$

where $e_S^{\mathrm{tot}}(x) = \int_{p_S^*(x)}^1 e_S(x,p)\, dp$. Skilled employment by quality, on the $d = 0$ region, balances inflows from unemployment, quality redraws, and ladder moves against outflows:

$$\begin{aligned}\big(\nu + \lambda_S + s^*(x,p)\, \kappa_S(1 - \Gamma(p))\big)\, e_S(x,p) &= \kappa_S\, u_S(x)\, \gamma(p) + \lambda_S\, \gamma(p) \int_{p_S^*(x)}^1 e_S(x,p')\, dp' \\ &\quad + \kappa_S\, \gamma(p) \int_{p_S^*(x)}^p s^*(x,p')\, e_S(x,p')\, dp', \qquad p \ge p_S^*(x). \end{aligned} \tag{44}$$

On the $d = 1$ region no skilled matches form, so $e_S(x,p) = 0$ and $u_S(x) = \phi\, t(x)/(\nu + f_U)$. The trained mass identity $m_S(x) = \phi\, t(x)/(\nu + d(x) f_U)$ follows from adding (43) to the integral of (44) over p. Untrained employment is the residual $e_U(x) = \ell(x) - m_S(x) - u_U(x) - t(x)$; it pools original untrained workers with ex-trained workers absorbed through the cross-market channel, which is what lets separations of the latter appear in (42) without a separate state variable.

A.3 Proof sketches: Propositions 1 and 2

Proposition 1 (training cutoff). $\Delta_T(x) = -c(x) + T(x) - U_U^{\mathrm{search}}(x)$. The cost term contributes $-c'(x) > 0$. From (4), $T'(x) = \phi\, U_S'(x)/(r + \phi + \nu) \ge 0$, since $U_S(x)$ is non-decreasing: both branches of (20) are non-decreasing in x ($\mathcal{I}_S$ rises with x under $\gamma_{P_S} \ge 1$; $E_U(x,1)$ rises with x), and the max of non-decreasing functions is non-decreasing. From (35), $U_U^{\mathrm{search}\prime}(x) = \beta_U f_U S_U'(x,1)/(r + \nu)$, with $S_U'(x,1)$ bounded by $P_U/(r + \nu + \delta_U)$-type terms. Monotone incentives—the requirement that $-c'(x) + T'(x)$ dominate $U_U^{\mathrm{search}\prime}(x)$—holds whenever the skilled technology is at least as steep in x as the unskilled one, which $\gamma_{P_S} \ge 1$ delivers alongside the strictly decreasing cost. Then Δ_T is increasing, crosses zero at most once, and the policy is the stated cutoff rule. $\square$

Proposition 2 (search-market cutoff). From (18)–(19),

$$\frac{d}{dx}\Big[U_S^{(1)}(x) - U_S^{(0)}(x)\Big] = \frac{f_U\, E_U'(x,1)}{r+\nu+f_U} - \frac{\kappa_S \beta_S\, \mathcal{I}_S'(x)}{r+\nu}.$$

$E_U(x,1)$ inherits the linear unskilled technology, so $E_U'(x,1)$ is bounded; $\mathcal{I}_S'(x)$ scales with $\gamma_{P_S} x^{\gamma_{P_S}-1}$ and dominates for x away from zero when $\gamma_{P_S} > 1$. In the parameter region of interest (skill-biased P_S, interior p_S^*) the difference is decreasing in x, so the $d=1$ region is a lower interval $[0,\underline{x}]$ with the indifference condition of the text at $\underline{x}$. □

A.4 Proof of Proposition 3

Take the skilled market on the $d=0$, no-OJS region, holding κ_S fixed. Write $\pi \equiv \gamma_{P_S} x^{\gamma_{P_S}}$, $q \equiv q_S^a(x)$, $\delta \equiv \delta_S(x)$, and define

$$A \equiv \frac{\partial\,[(r+\nu)U_S(x)]}{\partial b_S}, \qquad B \equiv \frac{\partial\, \mathcal{I}_S(x)}{\partial b_S}.$$

From the surplus recursion (11), $\partial S_S/\partial b_S = (-A+\lambda_S B)/(r+\nu+\lambda_S)$, independent of p. Differentiating $\mathcal{I}_S = \int_{p_S^*}^1 S_S\, d\Gamma$ and using $S_S(x,p_S^*)=0$ (the Leibniz boundary term vanishes),

$$B = q\,\frac{-A+\lambda_S B}{r+\nu+\lambda_S} \quad\Longrightarrow\quad B = \frac{-q\,A}{r+\nu+\delta},$$

where the second equality uses $r+\nu+\lambda_S-\lambda_S q = r+\nu+\delta$. From the unemployment value (18), $A = 1+\kappa_S\beta_S B$, so

$$A = \frac{r+\nu+\delta}{r+\nu+\delta+\kappa_S\beta_S\, q} \in (0,1).$$

Finally, differentiating the reservation rule $p_S^* = [(r+\nu)U_S - \lambda_S \mathcal{I}_S]/\pi$,

$$\frac{\partial p_S^*}{\partial b_S} = \frac{A-\lambda_S B}{\pi} = \frac{A\,(r+\nu+\delta+\lambda_S q)}{(r+\nu+\delta)\,\pi} = \frac{r+\nu+\lambda_S}{\big[r+\nu+\delta+\kappa_S\beta_S\, q\big]\,\pi},$$

which is (28); positivity is immediate and the response is decreasing in x through π (and through δ, q where p_S^* falls with x). The unskilled derivation is identical with $(\lambda_U, G, f_U, \beta_U, P_U x)$ in place of $(\lambda_S, \Gamma, \kappa_S, \beta_S, \pi)$ and acceptance probability 1, starting from (35); it yields

$$A_U = \frac{r+\nu+\delta_U}{r+\nu+\delta_U+f_U\beta_U}, \qquad \frac{\partial p_U^*}{\partial b_U} = \frac{r+\nu+\lambda_U}{\big[r+\nu+\delta_U+f_U\beta_U\big]\,P_U\,x}. \qquad \square$$

A.5 Proof of Proposition 4

All derivatives hold tightness fixed and evaluate each branch at its own fixed point.

(i) Effects of b_U. The training margin: b_U enters $\Delta_T(x) = -c(x) + T(x) - U_U^{\text{search}}(x)$ only through U_U^{search}, and from the unskilled block of Appendix A.4, $\partial U_U^{\text{search}}/\partial b_U = A_U/(r+\nu) > 0$. Hence $\partial \Delta_T/\partial b_U < 0$ pointwise; since Δ_T is increasing in x with $\Delta_T(\bar{x}) = 0$, the cutoff rises. The cross-market margin: b_U enters $U_S^{(1)}$ through $E_U(x,1) = U_U(x) + \beta_U S_U(x,1)$, with

$$\frac{\partial E_U(x,1)}{\partial b_U} = A_U \left[\frac{1}{r+\nu} - \frac{\beta_U}{r+\nu+\delta_U} \right] > 0,$$

using $\partial S_U(x,1)/\partial b_U = -A_U/(r+\nu+\delta_U)$ from Appendix A.4. $U_S^{(0)}$ does not involve b_U. Hence $U_S^{(1)} - U_S^{(0)}$ shifts up; since it is decreasing in x (Proposition 2), the crossing $\underline{x}$ moves right.

(ii) Effects of b_S. The training margin: b_S enters Δ_T only through $T(x) = (b_T + \phi U_S(x))/(r+\phi+\nu)$, and $\partial U_S/\partial b_S = A/(r+\nu) > 0$ on the relevant ($d=0$) region; hence $\partial \Delta_T/\partial b_S > 0$ and $\bar{x}$ falls. The cross-market margin:

$$\frac{\partial}{\partial b_S} \left[U_S^{(0)} - U_S^{(1)} \right] = \frac{A}{r+\nu} - \frac{1}{r+\nu+f_U},$$

which is positive iff $A\,(r+\nu+f_U) > r+\nu$. Substituting A from Appendix A.4 and simplifying,

$$A\,(r+\nu+f_U) > r+\nu \iff f_U\,(r+\nu+\delta_S) > \beta_S\,\kappa_S\,q_S^a\,(r+\nu),$$

the condition stated in the proposition. Under it, $U_S^{(0)} - U_S^{(1)}$ shifts up and $\underline{x}$ moves left. The economics of the condition: the $d=1$ worker expects to exit unemployment at rate f_U and therefore capitalises a b_S increase over a shorter horizon; only if her exit rate were very low relative to the (worker-share-weighted) skilled acceptance hazard could the ranking flip. □

A.6 Proof of Proposition 5

(i) Marginal equivalence. Hold U_S (hence T beyond its direct b_T-dependence) and U_U^{search} fixed; both policies enter Δ_T only directly. From (4), $\partial T/\partial b_T = 1/(r+\phi+\nu)$, so $\partial \Delta_T/\partial b_T = 1/(r+\phi+\nu)$. The cost function satisfies $\partial c(x)/\partial c = c(x)$, so a parameter change dc changes the cost *level* at the margin by $c(\bar{x})\,dc$ and $\partial\Delta_T/\partial c = -c(x)$. The implied cutoff shifts are

$$d\bar{x}\big|_{b_T} = -\frac{db_T/(r+\phi+\nu)}{\Delta_T'(\bar{x})}, \qquad d\bar{x}\big|_{c} = \frac{c(\bar{x})\,dc}{\Delta_T'(\bar{x})},$$

which coincide when $c(\bar{x})\,|dc| = db_T/(r+\phi+\nu)$: the stipend equals an up-front transfer of its expected discounted value over the spell.

(ii) Fiscal flow ratio. In stationary equilibrium the training stock is $M_T = \int t = E_f/(\phi+\nu)$, where $E_f = \int_{\bar{x}}^1 \tau(x)u_U(x)\,dx$ is the entry flow. The stipend's fiscal flow is $db_T \cdot M_T = db_T\,E_f/(\phi+\nu)$. The proportional cost subsidy reimburses $c(x)\,|dc|$ to each entrant of type x, a fiscal flow of $|dc|\int_{\bar{x}}^1 c(x)\,\tau(x)u_U(x)\,dx = |dc|\;\bar{c}_e\,E_f$. Taking the ratio and substituting the equivalence $|dc| = db_T/[(r+\phi+\nu)\,c(\bar{x})]$,

$$\frac{\text{stipend}}{\text{cost subsidy}} = \frac{db_T\,E_f/(\phi+\nu)}{|dc|\,\bar{c}_e\,E_f} = \frac{r+\phi+\nu}{\phi+\nu}\cdot\frac{c(\bar{x})}{\bar{c}_e}.$$

Since c is strictly decreasing and every entrant has $x > \bar{x}$, $\bar{c}_e < c(\bar{x})$; and $r > 0$ makes the first factor exceed one. □

Remark. Both parts are partial-equilibrium statements at the margin: they hold U_S, U_U^{search}, and tightness fixed and consider small reforms. The quantitative exercises of Section 6.2 relax all three.

B Computation

The stationary equilibrium is computed as two market blocks wrapped in a global fixed point on the link variables. The unskilled block consumes the skilled unemployment value $U_S(\cdot)$ and the cross-market seeker density $d(\cdot)u_S(\cdot)$; the skilled block consumes the trained mass $m_S(\cdot)$, the unskilled contact rate f_U, and the fresh-match value $E_U(\cdot,1)$. Type and quality live on fixed grids; all integrals against G and Γ are precomputed quadratures.

B.1 Unskilled block

Given $U_S(\cdot)$ and $d(\cdot)u_S(\cdot)$, the block iterates on the outer unknowns $(\theta_U, p_U^*(\cdot), \tau(\cdot), u_U(\cdot), t(\cdot))$:

(1) **Initialise** $\theta_U^{(0)} > 0$, a feasible $p_U^{*,(0)}(\cdot) \in (0,1)$, a training policy $\tau^{(0)}$, and densities.

(2) **Inner value iteration.** Holding $(\theta_U^{(n)}, p_U^{*,(n)})$ fixed, cycle until convergence: (2.1) update T from (4) given U_S; (2.2) update $U_U = \max\{U_U^{\text{search}}, -c+T\}$ and the implied

τ; (2.3) solve the match-value system (32)–(33) with the Nash split, given U_U; (2.4) update U_U^{search} from (6) with $f_U^{(n)} = \theta_U^{(n)} q_U(\theta_U^{(n)})$.

(3) **Distributions.** Solve (41)–(42) pointwise in x for (u_U, t), with e_U as the residual mass.

(4) **Reservation update.** Recompute $p_U^*(\cdot)$ from the closed form (8); apply rank-1 Anderson acceleration to the grid vector before writing back.

(5) **Tightness update.** Update θ_U from free entry (10) on the augmented pool $u_U + d\, u_S$, inverting $q_U(\theta) = \mu_U \theta^{-\eta_U}$; Anderson-accelerate.

(6) **Converge** on $\max\{|\Delta\theta_U|, \|\Delta p_U^*\|_\infty, \|\Delta u_U\|_\infty\}$, else return to (2).

B.2 Skilled block

Given $(m_S, f_U, E_U(\cdot, 1))$, the block iterates on $(\theta_S, p_S^*(\cdot), p_S^{oj}(\cdot))$:

(1) **Initialise** $\theta_S^{(0)}$ and feasible cutoffs.

(2) **Inner surplus iteration.** Holding the outer unknowns fixed, with $\kappa_S = \theta_S^{(n)} q_S(\theta_S^{(n)})$, cycle until convergence: (2.1) accumulate the tail integral $\mathcal{I}_S(x) = \int_{p_S^*}^1 \max\{S_S^0, S_S^1\}\, d\Gamma$; (2.2) compute the two branches $U_S^{(0)}, U_S^{(1)}$ from (18)–(19) and set $d(x)$, $U_S(x)$ by (20)—the discrete choice is a by-product of the value iteration and needs no separate loop; (2.3) update S_S^0, S_S^1 on the quality grid from (11)–(12); (2.4) apply the Nash split to recover (E_S^s, J_S^s).

(3) **Cutoff updates.** Set $p_S^*(x)$ at the first grid point where $\max\{S_S^0, S_S^1\} \geq 0$ and $p_S^{oj}(x)$ at the first where $S_S^1 - S_S^0 \leq 0$; damp both updates.

(4) **Distributions.** For $d(x) = 0$ types, solve (43)–(44) as a linear system in $\{e_S(x, p_i)\}_i$ per type; for $d(x) = 1$ types use the closed forms $u_S = \phi t/(\nu + f_U)$, $e_S \equiv 0$. Form the seeker density $\widetilde{u}_S$ of (16).

(5) **Tightness update.** Compute the contact-value integral $\mathcal{J}_S$ and update θ_S from $q_S(\theta_S) = k_S/\mathcal{J}_S$; Anderson-accelerate.

(6) **Converge** on $\max\{|\Delta\theta_S|, \|\Delta p_S^*\|_\infty, \|\Delta p_S^{oj}\|_\infty\}$, else return to (2).

B.3 Global iteration

(1) Initialise $U_S^{(0)}(\cdot)$, $m_S^{(0)}(\cdot)$, and $d^{(0)} u_S^{(0)} \equiv 0$ (the first pass is the baseline no-cross-market solve).

(2) Solve the unskilled block; export f_U, $E_U(\cdot, 1)$, $t(\cdot)$.

(3) Solve the skilled block; obtain U_S^{new}, d^{new}, u_S^{new}.

(4) Update $m_S^{\text{new}}(x) = \phi\, t(x)/(\nu + d^{\text{new}}(x) f_U)$; apply Anderson acceleration (rank 1) to the joint state (U_S, m_S); write back $d^{\text{new}} u_S^{\text{new}}$.

(5) Stop when $\max\{\|\Delta U_S\|_\infty, \|\Delta m_S\|_\infty\} < \text{tol}$, else return to (2).

Only (U_S, m_S) are accelerated; the remaining link objects are recomputed deterministically each pass. The surplus-only reduction keeps the state small: wages never enter the solver and are recovered afterwards from the closed forms (9), (14)–(15).

B.4 Transition path

The transition between two stationary equilibria is a forward–backward fixed point on the tightness paths:

(0) **Boundary objects.** Solve the pre- and post-switch stationary equilibria.

(1) **Guess** paths $\{\theta_U^{(0)}(t_n), \theta_S^{(0)}(t_n)\}_{n=0}^{N}$ on a time grid spanning the switch date to a horizon at which the post-switch steady state binds.

(2) **Backward pass.** Given the tightness paths, solve the time-dependent value system backward from the terminal condition (post-switch steady-state values). At each date compute the two U_S branches, set $d(x,t)$, and propagate the surplus recursions; recover the policy paths $(\tau, p_U^*, p_S^*, p_S^{oj}, d)$.

(3) **Forward pass.** Given the policy paths, integrate the laws of motion (23)–(25) (and the e_S analogue) forward from the pre-switch stationary distributions.

(4) **Tightness update.** Recompute $\{\theta_U(t_n), \theta_S(t_n)\}$ from free entry date by date, on the augmented unskilled pool and the seeker-corrected skilled pool; damp.

(5) **Converge** on the sup-norm change of both tightness paths, else return to (2).

Along the path, all hazards and policies for $t > T$ are those implied by the post-switch parameters; the distributions at T^+ equal the pre-switch stationary ones. In a tranquil pre-switch regime $d^{\text{pre}} \equiv 0$, so the initial condition coincides with the baseline state, and any activation of the cross-market channel is an equilibrium outcome of the post-switch system, not an assumption.

C Data and Estimation Details

C.1 Samples and moment construction

Common restrictions. Civilians aged 16–64. Wage moments use wage and salary workers with positive income, weeks, and hours, excluding the self-employed; hourly wages are deflated and normalised as described below. Skill classification: skilled = $\mathbf{1}\{\text{EDUC} \geq \text{bachelor's}\}$ (IPUMS code 111). The training indicator marks college enrolment without a completed BA.

Stocks (CPS Basic). Unemployment rates and the skilled share are computed on the training-excluded labour force, so that the model's three-pool partition (unskilled LF, skilled LF, trainees) maps cleanly onto the data. The training share is the share of full-time students—enrolled without a BA and out of the labour force—in the working-age population, rescaled by a window-specific factor κ_w that anchors the CPS level to the NSC/IPEDS enrolment universe; the CPS enrolment flags alone miss part-time and non-degree enrolment.

Transitions (CPS Basic matched panel). Monthly transitions match individuals on CPSIDP across consecutive months within rotation spells (months-in-sample $\{1, 2, 3, 5, 6, 7\}$ matched forward). Job-finding is $\Pr(E_{t+1} \mid U_t)$ and separation $\Pr(U_{t+1} \mid E_t)$, by skill. The training-entry hazard is the monthly probability that an unskilled, unemployed, not-enrolled worker is enrolled the following month. For March–June 2020, the BLS misclassification of absent employed workers is corrected by reassigning "employed, absent for other reasons" to unemployment, following the BLS guidance.

Wages (ASEC). Real hourly wages by skill group; the windows map to ASEC survey years shifted one year forward, since ASEC reports prior-year income (e.g. the 2020–22 crisis window uses ASEC 2021–23). Medians and within-group variances are computed on normalised wages; the log skill premium is $E[\log w \mid S] - E[\log w \mid U]$.

Tightness (JOLTS + CPS). JOLTS vacancies are allocated to skill segments through the CPS industry-by-education employment composition at the two-digit level, then divided by the segment unemployment stocks. The skilled EE rate comes from Census J2J (E4 flows: employer-to-employer hires), restricted to the skilled education group.

C.2 Pre-estimated parameters

Completion rate ϕ. In steady state the training share approximately equals the entry flow over $\phi + \nu$, so ϕ and the cutoff $\bar{x}$ are not separately identified from CPS stocks and noisy monthly enrolment flows. I therefore compute ϕ from administrative data: average IPEDS-universe enrolment in 4-year and 2-year institutions ($E_{4\text{yr}}, E_{2\text{yr}}$, pooled over Fall semesters 2003–2024) and NCES median times-to-degree ($d_{4\text{yr}} = 49$ months, $d_{2\text{yr}} = 37$ months), combined as the enrolment-weighted completion hazard

$$\hat{\phi} = \frac{E_{4\text{yr}}/d_{4\text{yr}} + E_{2\text{yr}}/d_{2\text{yr}}}{E_{4\text{yr}} + E_{2\text{yr}}} = 0.0222 \text{ per month.}$$

Turnover rate ν. The turnover rate enters value functions as $r+\nu$ and is nearly collinear with r in the SMM objective, so it is pre-estimated by a life-table method on the CPS labour force: for each participant of age a, remaining working life is $(65 - a) \times 12$ months, and $\hat{\nu}_w$ is the reciprocal of its weighted mean over baseline window w. This yields $\hat{\nu} = 0.00323$ per month on the pre-FC baseline and 0.00336 on the pre-COVID baseline; each crisis window inherits its baseline's value. The discount rate is calibrated at $r = 0.0033$ per month ($\approx 4\%$ annually); its near-collinearity with the flow payoffs and productivity shifters is the reason it is not estimated.

C.3 Influence functions and the moment covariance matrix

For each window w, every moment is expressed as an asymptotically linear statistic, $\hat{m}_k = N^{-1}\sum_i \psi_k(z_i) + o_p(N^{-1/2})$, and the covariance matrix is formed from the stacked influence functions, $\hat{\Sigma}_w = N_w^{-1}\sum_i \boldsymbol{\psi}(z_i)\boldsymbol{\psi}(z_i)'$. The forms used: $\psi_i = y_i - \hat{m}$ for means; $\psi_i = (y_i - \bar{y})^2 - \hat{\sigma}^2$ for variances; $\psi_i = (y_i - \hat{r}\, x_i)/\bar{x}$ for ratios and rates (unemployment rates, shares, all transition hazards, tightness); $\psi_i = (\mathbf{1}\{y_i \leq \hat{q}_\alpha\} - \alpha)/f(\hat{q}_\alpha)$ for quantiles. Contributions are stacked at the period level before the outer product, so within-period clustering is respected. The κ_w rescaling of the training share is applied to $\hat{\Sigma}_w$ at load time by scaling the corresponding row and column (the diagonal cell by κ_w^2), leaving the covariance structure of the remaining moments untouched.

C.4 Weighting and optimisation

Moment deviations are scale-normalised by the absolute value of the data moment, $\tilde{g}_k(\vartheta) = (\hat{m}_k - m_k(\vartheta))/|\hat{m}_k|$, so that rates, shares, and tightness ratios contribute comparably to the objective. The headline estimation weights the normalised deviations equally ($W = I$). Two robustness configurations use the stored $\hat{\Sigma}_w$: diagonal weights $W = \mathrm{diag}(\hat{\Sigma}_w)^{-1}$, and the full optimal matrix $W = \hat{\Sigma}_w^{-1}$ with off-diagonal shrinkage,

$$\hat{\Sigma}(\alpha) = \Sigma_D + (1-\alpha)\Sigma_{\mathrm{od}}, \qquad \hat{\alpha} = \inf\{\alpha \in [0,1] : \kappa(\hat{\Sigma}(\alpha)) \leq \kappa^*\},$$

applied when the condition number $\kappa(\cdot)$ of the raw matrix exceeds the target κ^*; with $K = 18$ moments and 36–60 monthly observations per window, the required shrinkage is modest for the baselines and larger for the crisis windows.

The SMM objective is minimised with a surrogate-based global optimiser (adaptively scaled radial basis functions over the parameter box, with local polishing of incumbent optima); each objective evaluation solves the stationary equilibrium of Appendix B at the candidate parameter vector. Standard errors for the structural parameters are computed by the bootstrap over the influence-function distribution of the data moments. **[TODO: Confirm final optimiser settings and SE protocol once the re-estimation completes.]**